

A Constructive Orbit-Based Theory of Learnability

Anonymous authors

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Abstract

We propose a structural theory of learnability from an orbit-based operator-theoretic perspective in which learnability is organized by dataset-induced orbit geometry and sparse constructive realizability rather than by unrestricted parametrization alone. We develop this perspective in finite dimensions, while showing that its core constructive mechanisms already persist in infinite-dimensional Banach settings.

operator-theoretic machine learning, orbit-based learning, weighted backward shifts, constructive realizability, sparse approximation

1 Introduction

Deep learning excels across vision, language, and time series, but much of its practical success still relies on stacked nonlinearities, heavy overparameterization, and architectural heuristics such as attention, which often obscure mechanism and demand large labeled datasets and careful tuning. In this work we ask a different question: can learning be organized around simple orbit dynamics, explicit geometry, and constructive realization rather than around increasingly opaque parametrization? We introduce an orbit-based framework in which scaffold geometry, realizability, and prediction are treated as three linked structural bottlenecks rather than as disconnected stages of a pipeline. The predictive instantiation of this framework, which we call a Weighted Backward Shift Neural Network (WBSNN), has a general-purpose design: the same orbit-based mechanism can be deployed across varied and demanding settings without requiring the pipeline to be tailored to each dataset. Moreover, although the backbone is linear, the model’s predictions become nonlinear through their dependence on orbit evolution and data-dependent coefficient selection, yielding both interpretability and expressive capacity. Empirically, WBSNN often recovers the orbit scaffold from just 1% of the training data while matching or surpassing strong baselines, offering a data-efficient and structurally lightweight alternative.

The philosophy is the following. Instead of taking the model as the primary object and asking it to discover structure implicitly, we first extract from the data a structured orbit scaffold encoding part of the dataset’s intrinsic geometry. On that scaffold we study whether a target operator, or in the predictive setting a target dataset map, can be constructively realized with explicit and analyzable error. Prediction is then built on top of this constructive stage: rather than learning outputs freely, the predictor learns a coefficient law that tracks and extends the constructive rule found on the training set. This leads to a framework organized around three explicit bottlenecks: a *scaffold bottleneck*, measuring how well the selected orbit scaffold captures the relevant geometry of the data; a *realizability bottleneck*, measuring how well the projected target can be constructively synthesized on that scaffold; and a *predictive coefficient bottleneck*, measuring how well the learned predictor reproduces and generalizes the constructive coefficient rule.

The theoretical contribution of the paper has two tightly connected parts. On the realizability side, we prove exact interpolation theorems showing that orbit-generated realizations exist under explicit structural conditions; we derive a two-term decomposition separating projection and realization errors; we develop an algorithm that searches for an optimal orbit-constrained scaffold; and we introduce CCDS, a constructive deviation-controlled synthesis mechanism that yields explicit certificates and recovers classical greedy pursuit as a boundary case. On the predictive side, we show that the same structural logic carries over to learning: the WBSNN predictor inherits the scaffold and realizability bottlenecks and adds only one new one, namely

the learned approximation of the effective CCDS coefficient map. This yields in-sample structural prediction bounds and an out-of-sample transfer principle under geometric proximity assumptions between training and test points. In summary, the paper should be read as a single theory of learnability: Section 2 develops its constructive realizability backbone, and Section 3 shows that prediction is obtained by extending the same scaffold-realization logic rather than by introducing a separate model class.

Why should the machine-learning community care about such a framework? Because it shifts the focus of learnability away from opaque feature mixing and unrestricted parametrization, and toward a more explicit structural question: what geometry the dataset induces, what targets can be constructively realized on that geometry, and how prediction can be built by extending that constructive rule. In this sense, the framework is structurally interpretable at the realization level: the scaffold is data-induced and geometric, the realization step is explicit and constructive, and every prediction is decomposed through coefficients over a fixed orbit-derived representation. The constructive theory also yields two properties of direct practical value. First, CCDS supplies explicit sparsity certificates, and the experiments confirm that accurate prediction is supported by highly sparse effective realizations. Second, the theory shows that exact block interpolation, while canonical, is not required for robust prediction. The framework therefore justifies the cheaper, more robust under noisy local blocks and better-conditioned relaxed regularized interpolant as the practical default, with predictive performance governed by scaffold geometry invariance rather than machine-precision interpolation.

More fundamentally, the framework suggests that datasets should not be viewed merely as collections of samples to be fit, but as objects carrying intrinsic geometric structure that directly shapes what can be learned from them.

Our construction is rooted in the theory of linear dynamics, where the weighted backward shift operator plays a foundational role. In infinite-dimensional separable sequence Banach spaces, this operator acts on sequences $x = (x_0, x_1, x_2, \dots)$ by shifting coordinates backward and scaling by a bounded weight sequence $w = (w_i)_{i \geq 0}$:

$$B_w x = (w_1 x_1, w_2 x_2, w_3 x_3, \dots). \quad (1)$$

The orbit of a point x under B_w is the sequence $\text{Orb}(x, B_w) = \{B_w^m x : m \geq 0\}$, that is, the collection of all iterates generated by repeatedly applying the shift. This simple mechanism gives rise to rich dynamical behavior, including dense orbits and hypercyclicity, and is therefore a natural source of structured representations. To adapt this dynamics to finite-dimensional learning problems, we introduce a wrapped finite-dimensional surrogate that cyclically reintroduces lost coordinates while preserving the shift-based orbit structure. This weighted backward shift mechanism serves as the backbone of the entire framework: it generates the orbit dictionary used to build scaffolds, supports the exact interpolation theory underlying realizability, and provides the structured representation class on which WBSNN performs prediction.

In this sense, the paper shifts the center of gravity of learning theory from parametrization alone to the interaction between dataset geometry, orbit structure, and constructive realization.

Relation to prior work. Our deviation-control viewpoint is motivated by Bernstein lethargy theory, where approximation errors along nested families need not obey a universal decay rate (Bernstein, 1954; Shapiro, 1964). Rather than postulating such a rate, we treat the residual trajectory itself as a constructive object. This differs from discrepancy principles, where a prescribed residual typically serves as a stopping or regularization criterion (Morozov, 1966; Bakushinsky & Kokurin, 2004), and from classical greedy pursuits such as MP and OMP, whose residuals are determined by greedy contraction or orthogonal projection (Mallat & Zhang, 1993; Pati et al., 1993; Tropp & Gilbert, 2007; DeVore & Temlyakov, 1996; Temlyakov, 2011). In our framework, feasible residual profiles are prescribed and realized explicitly; standard greedy updates arise as boundary regimes of this broader deviation-control mechanism.

Operator-theoretic ideas in machine learning also appear in Koopman-based representations and forecasting (Mezić, 2005; Lusch et al., 2018; Azencot et al., 2020; Brunton et al., 2022) and in neural-operator approaches such as DeepONet and neural integral equations (Lu et al., 2021; Zappala et al., 2024). WBSNN is different in that it does not learn an operator representation directly: iterates of a single orbit-generating operator first induce a data-dependent scaffold, targets are then constructively realized on that scaffold, and

prediction is obtained by extending the resulting coefficient law. Its mathematical origin is therefore closer to linear dynamics and orbit theory, with weighted backward shifts providing the guiding prototype (Bayart & Matheron, 2009).

The framework’s backbone. Weighted backward shifts are classical operators in linear dynamics and can exhibit remarkably rich orbit behavior in infinite-dimensional spaces. To adapt their orbit mechanism to finite-dimensional learning, we use a wrapped weighted backward shift, which replaces coordinate loss by cyclic propagation modulo d . This provides the structured orbit backbone used throughout the framework.

Definition 1.1 (The wrapped weighted backward shift). *A wrapped weighted backward shift $W : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is defined as $We_j = w_j e_{j-1 \pmod d}$, where $e_0, \dots, e_{d-1}$ is the canonical basis and $w_j \neq 0$.*

Lemma 1.2. *For every $m \geq 1$ and $z \in \mathbb{R}^d$,*

$$(W^m z)_{i-1} = \left(\prod_{r=0}^{m-1} w_{(i+r) \pmod d} \right) z_{(i+m-1) \pmod d}, \quad i = 1, \dots, d.$$

The proof is deferred to Appendix A. For intuition, the reader may view $W^L x$ as a shift-and-scale transformation of x , beginning at the L -th entry and preserving the cyclic order of the components modulo d .

Set $\lambda := \prod_{j=0}^{d-1} w_j$. Since every coordinate crosses every weight exactly once in d iterates, $W^d = \lambda I_{\mathbb{R}^d}$. Hence:

Lemma 1.3. *If $L = md + r$, $0 \leq r < d$, then $W^L = \lambda^m W^r$.*

2 Orbit-Based Constructive Realization of Linear Operators

This section develops the constructive backbone of the paper. Let $\Xi = \{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^C$ be a finite input–target dataset and let $A : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be a linear target operator. We use the empirical mean seminorm $\|A\|_{\Xi} := \frac{1}{n} \sum_{i=1}^n \|A(x_i)\|_2$.

The central object is a data-induced orbit scaffold $U \subseteq \mathbb{R}^C$ constructed from a small anchor subset of Ξ . The target is first projected onto this scaffold and then constructively realized within it. This separates operator realization into two distinct problems: how well the scaffold captures the target on the dataset, and how accurately the projected target can be synthesized once the scaffold is fixed. These will become, respectively, the projection and realization bottlenecks developed below.

Standing data notation. Throughout, let

$$\Xi_{\text{train}} = \{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^C, \quad \Xi_{\text{sub}} \subseteq \Xi_{\text{train}},$$

where Ξ_{sub} is the anchor subset used for scaffold construction. We write

$$\Xi_{\text{sub}} = D_1 \sqcup \dots \sqcup D_K, \quad D_k = \{(x_{k,j}, y_{k,j})\}_{j=1}^{s_k}.$$

When out-of-sample prediction is considered, we additionally write

$$\Xi_{\text{test}} = \{(\bar{x}_j, \bar{y}_j)\}_{j=1}^m.$$

add A linear, xi norm

2.1 Blockwise orbit interpolation

For each $1 \leq k \leq K$, choose depths $L_{k,1}, \dots, L_{k,s_k} \in \{0, \dots, d-1\}$ such that the orbit representatives $z_{k,j} := W^{L_{k,j}} x_{k,j}$, $j = 1, \dots, s_k$, are linearly independent. Define

$$Z_k := [z_{k,1} \quad \dots \quad z_{k,s_k}] \in \mathbb{R}^{d \times s_k}, \quad Y_k := [y_{k,1} \quad \dots \quad y_{k,s_k}] \in \mathbb{R}^{C \times s_k}. \quad (2)$$

Since Z_k has full column rank, the canonical interpolant $J_k := Y_k Z_k^\dagger \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^C)$ satisfies

$$J_k W^{L_{k,j}} x_{k,j} = J_k z_{k,j} = y_{k,j}, \quad j = 1, \dots, s_k. \quad (3)$$

Moreover, $\|J_k\| \leq \|Y_k\|_F / \sigma_{\min}(Z_k)$, so the conditioning of the orbit matrix Z_k controls the amplification associated with blockwise interpolation.

Exact interpolation also survives in the infinite-dimensional setting through a constructive application of the Hahn–Banach theorem; readers interested in this extension are referred to Appendix A.1.

A natural question is how restrictive the orbit-independence hypothesis underlying exact interpolation is. The next result shows that, whenever this condition is feasible for a given block, it is structurally non-fragile: the admissible weights form an open dense subset of weight space, and the exceptional set has Lebesgue measure zero. The proof uses only finite-dimensional polynomial algebra and is deferred to Appendix A.1.

Theorem 2.1 (Genericity of the orbit-independence condition). *Let $D = (x_i, y_i)_{i=1}^s \subset \mathbb{R}^d \times \mathbb{R}^C$, $s \leq d$, and let $\Omega := (\mathbb{R} \setminus 0)^d$. For each $w = (w_0, \dots, w_{d-1}) \in \Omega$, let W_w denote the wrapped weighted backward shift determined by w . Define*

$$\Omega_D := \left\{ w \in \Omega : \exists L_1, \dots, L_s \in \{0, \dots, d-1\} \text{ such that } W_w^{L_i} x_i \notin \text{span}\{W_w^{L_1} x_1, \dots, W_w^{L_{i-1}} x_{i-1}\}, 1 \leq i \leq s \right\}.$$

If $\Omega_D \neq \emptyset$, then Ω_D is open and dense in Ω . Moreover, $\Omega \setminus \Omega_D$ has Lebesgue measure zero.

From local interpolation to the orbit scaffold. Equation (3) selects one orbit depth $L_{k,j}$ for each anchor in order to interpolate its target exactly. The scaffold, however, is built from the full finite orbit family generated by these anchors. Specifically, define the orbit-based dictionary

$$\mathcal{D} := \left\{ J_k W^m x_{k,j} : (x_{k,j}, y_{k,j}) \in D_k, 1 \leq k \leq K, 0 \leq m \leq d-1 \right\} \subseteq \mathbb{R}^C. \quad (4)$$

We use this notation throughout:

$$\mathcal{I} := \{(k, j, m) : 1 \leq k \leq K, 1 \leq j \leq s_k, 0 \leq m \leq d-1\}.$$

We call $U := \text{span } \mathcal{D}$ the orbit scaffold and denote $\Pi_U : \mathbb{R}^C \rightarrow \mathbb{R}^C$ the associated orthogonal projector. When CCDS requires unit-norm atoms, $q_{k,j,m}$ denotes the normalized atom associated with $J_k W^m x_{k,j}$.

By Lemma 1.3, depths $0 \leq m \leq d-1$ contain all distinct orbit directions generated by W up to scalar multiplication, since every deeper iterate is a scalar multiple of one of these residue classes. The selected depths $L_{k,j}$ guarantee exact interpolation on the anchors. Thus, $\{y : (x, y) \in \Xi_{\text{sub}}\} \subseteq \mathcal{D}$.

The two realizability bottlenecks. Once the scaffold U is fixed, any realization error can be separated by inserting the projected target $\Pi_U A$. The resulting two terms have different roles: the first measures whether the target itself is well represented by the selected scaffold, while the second measures whether that projected target can be constructively synthesized within the scaffold.

Theorem 2.2 (Two-term error decomposition for operator realization on an orbit scaffold). *Let $A : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be linear, and let $\hat{y} : \mathbb{R}^d \rightarrow \mathbb{R}^C$. Then*

$$\|A - \hat{y}\|_{\Xi_{\text{train}}} \leq \underbrace{\|A - \Pi_U A\|_{\Xi_{\text{train}}}}_{\varepsilon_{\text{proj}}} + \underbrace{\|\Pi_U A - \hat{y}\|_{\Xi_{\text{train}}}}_{\varepsilon_{\text{realiz}}}. \quad (5)$$

The decomposition identifies the two structural bottlenecks governing realization on an orbit scaffold. The projection error $\varepsilon_{\text{proj}}$ measures how well U captures the target set $\{Ax : (x, y) \in \Xi_{\text{train}}\}$, and depends only on the target operator and the selected scaffold. The realization error $\varepsilon_{\text{realiz}}$ measures how accurately the projected target $\Pi_U A$ is realized once U has been fixed.

The remainder of this section addresses these two bottlenecks in order: first the selection of an orbit scaffold with small projection error, and then the constructive realization of the projected target on that scaffold.

2.2 Collapsing the projection error: the scaffold selection problem.

We first show that the existence of optimal scaffolds under which projection error collapses, is a mathematically well-posed problem. Denote by $\mathcal{U}$ the set of all scaffolds relative to Ξ_{train} .

Theorem 2.3 (Existence of the best orbit-constrained scaffold). *Let $A : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be linear, then there exists $U_\star \in \mathcal{U}$ such that $\|A - \Pi_{U_\star} A\|_{\Xi_{\text{train}}} = \min_{U \in \mathcal{U}} \|A - \Pi_U A\|_{\Xi_{\text{train}}}$.*

Proof of Theorem 2.3 deferred to Appendix A.2.

From existence to computation: controlled scaffold expansion. Theorem 2.3 guarantees the existence of an optimal orbit-constrained scaffold, but it is nonconstructive. The remaining problem is algorithmic: how to enlarge a scaffold within the admissible orbit geometry so as to reduce the projection error. Unlike unconstrained low-rank approximation, the admissible subspaces here must arise from the orbit construction. We therefore study nested one-dimensional scaffold expansions and characterize the residual reductions they can achieve.

The analysis below identifies the exact interval of feasible one-step drops, gives an explicit construction for any admissible drop, and shows how these steps can be combined to realize prescribed feasible projection-error profiles. Greedy spectral selection appears as the maximal-drop endpoint of this construction.

RMS projection residual. For scaffold selection, we work with the empirical RMS projection residual, whose squared form provides the Hilbertian energy structure needed below. Let $A : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be linear, and define the target snapshot matrix $Y := [Ax_1, \dots, Ax_N] \in \mathbb{R}^{C \times N}$. For a scaffold $U \subset \mathbb{R}^C$, set $\varepsilon_{\text{proj}}^{\text{rms}}(A, U) := N^{-1/2} \|(I - \Pi_U)Y\|_F$. The mean and RMS projection errors satisfy $\varepsilon_{\text{proj}}(A, U) \leq \varepsilon_{\text{proj}}^{\text{rms}}(A, U) \leq \sqrt{N} \varepsilon_{\text{proj}}(A, U)$, by the standard relations between the ℓ^1 and ℓ^2 norms. Thus controlling the RMS residual also controls the projection bottleneck in Theorem 2.2.

One-step scaffold expansion. Fix an admissible scaffold space $V \subset \mathbb{R}^C$ and an admissible scaffold $U \subseteq V$. Define $S := V \cap U^\perp$, $R := (I - \Pi_U)Y$, $E := \|R\|_F^2$, and $M := \Pi_S R R^\top \Pi_S$.

Proposition 2.4 (One-step feasible scaffold expansion). *For every unit vector $u \in S$, let $U^+ := U \oplus \text{span}\{u\}$ and $E^+ := \|(I - \Pi_{U^+})Y\|_F^2$. Then $E^+ = E - q(u)$, $q(u) := \|u^\top R\|_2^2 = u^\top M u$. Define $\lambda_{\min} := \min_{\|u\|_2=1, u \in S} u^\top M u$, $\lambda_{\max} := \max_{\|u\|_2=1, u \in S} u^\top M u$. Then the achievable one-step drops are exactly $[\lambda_{\min}, \lambda_{\max}]$.*

Moreover, if $\dim S \geq 2$ and $\lambda_{\max} > \lambda_{\min}$, let $v_{\max}, v_{\min} \in S$ be unit eigenvectors associated with $\lambda_{\max}$ and $\lambda_{\min}$, respectively. For every prescribed drop $e \in [\lambda_{\min}, \lambda_{\max}]$, the unit vector

$$u_e := \sqrt{\frac{e - \lambda_{\min}}{\lambda_{\max} - \lambda_{\min}}} v_{\max} + \sqrt{\frac{\lambda_{\max} - e}{\lambda_{\max} - \lambda_{\min}}} v_{\min}$$

satisfies $q(u_e) = e$, and hence $E^+ = E - e$.

The proof is given in Appendix A.2. Proposition 2.4 gives both the exact feasibility interval for a one-step scaffold expansion and an explicit construction realizing any admissible interior drop.

Theorem 2.5 (Constructive deviation-control principle for scaffold selection). *Let $V \subset \mathbb{R}^C$ be an admissible scaffold space and let $Y \in \mathbb{R}^{C \times N}$ be fixed. Define $U_0 := \{0\}$, $R_0 := Y$, $E_0 := \|Y\|_F^2$. Fix an integer $T \geq 1$ and a target energy profile $E_0 \geq E_1 \geq \dots \geq E_T \geq 0$. For each $t = 1, \dots, T$, define $S_{t-1} := V \cap U_{t-1}^\perp$ and $M_{t-1} := \Pi_{S_{t-1}} R_{t-1} R_{t-1}^\top \Pi_{S_{t-1}}$. Assume $S_{t-1} \neq \{0\}$ and define $\lambda_{\min}^{(t-1)} := \min_{\|u\|_2=1, u \in S_{t-1}} u^\top M_{t-1} u$, $\lambda_{\max}^{(t-1)} := \max_{\|u\|_2=1, u \in S_{t-1}} u^\top M_{t-1} u$. If $E_{t-1} - E_t \in [\lambda_{\min}^{(t-1)}, \lambda_{\max}^{(t-1)}]$, $t = 1, \dots, T$, then there exist unit vectors $u_t \in S_{t-1}$*

and a nested scaffold chain $U_t := U_{t-1} \oplus \text{span}\{u_t\} \subseteq V$ such that, with $R_t := (I - \Pi_{U_t})Y$, one has $\|R_t\|_F^2 = E_t$, $t = 1, \dots, T$. Whenever $\dim S_{t-1} \geq 2$, the direction u_t can be chosen explicitly as in Proposition 2.4.

The proof is given in Appendix A.2. Conversely, if a prescribed drop lies outside the corresponding feasibility interval, no one-dimensional admissible scaffold expansion can realize that step.

Greedy SVD selection as the extremal case. The familiar greedy spectral update is recovered at the endpoint $e = \lambda_{\max}^{(t-1)}$. Indeed, the maximal one-step decrease is obtained by a leading eigenvector of M_{t-1} , equivalently by a leading left singular vector of $\Pi_{S_{t-1}} R_{t-1}$. This is the standard Rayleigh–Ritz/SVD choice. The deviation-control mechanism above is more general: rather than forcing maximal immediate contraction, it characterizes the full feasible interval and explicitly realizes any admissible interior drop, allowing an entire feasible residual profile to be prescribed step by step.

Transition to realizability and prediction. The scaffold-selection mechanism above controls the projection term in the two-term decomposition. Once an admissible scaffold has been selected, the remaining problem is constructive realization of the projected target on that scaffold, namely control of the realization term. We turn to this second bottleneck next. In the predictive setting of Section 3, the same structure persists: scaffold selection is followed by constructive realization and then by learning a coefficient law that extends this realization over the associated orbit dictionary.

2.3 Collapsing the realization error: the CCDS mechanism

Once U is fixed, standard sparse approximation methods such as Orthogonal Matching Pursuit (OMP) can be used to reduce the realization error, and in finite dimension can recover the orthogonal projection once a sufficient independent atom set has been selected. Our goal is therefore not to replace OMP merely as an approximation solver. Instead, we introduce a realization mechanism in which the residual trajectory itself becomes a controllable geometric object: Constructive Controlled Deviation Synthesis (CCDS).

Why CCDS if OMP already realizes the scaffold? OMP selects an atom, enlarges the current span, and refits by least squares; the resulting residual is therefore determined by the extremal orthogonal projection onto that span. CCDS retains the same span-building logic but replaces the rigid least-squares update by a controlled realization path: rather than accepting the residual dictated by the orthogonal projection at each step, it allows the residual decay to be prescribed within the geometry permitted by the current realization space. Thus, unlike OMP, which treats sparse realization as an endpoint approximation problem, CCDS turns it into a stepwise controllable process. Both OMP and CCDS can accurately realize targets on a fixed scaffold, but OMP is confined to the extremal least-squares update on each selected span. CCDS can instead move through the interior of the feasible deviation geometry by prescribing admissible residual levels, while OMP has no parameter with which to realize those prescribed interior paths. The difference appears in difficult geometry (such as weak or nearly degenerate dictionary geometry, see Experiment 4.1), where CCDS preserves this accuracy while maintaining a substantially more stable and controlled residual path. Thus deviation control adds robustness without sacrificing realizability.

From Bernstein-type prescribed approximation to deviation control. The CCDS viewpoint is motivated by the Bernstein-type lethargy theory question of prescribing approximation errors along a nested family of subspaces, rather than merely taking the best approximation available at each stage. Let

$$Y_0 \subset Y_1 \subset \cdots \subset Y_T \subset H, \quad Y_t = Y_{t-1} \oplus \text{span}\{u_t\}, \quad u_t \in Y_{t-1}^\perp, \quad \|u_t\| = 1.$$

Given a target $y \in H$, initialize $\hat{y}_0 = 0$ and $r_0 = y$, and update

$$\hat{y}_t = \hat{y}_{t-1} + \lambda_t u_t, \quad r_t = r_{t-1} - \lambda_t u_t. \quad (6)$$

Instead of forcing the extremal projection at every step, prescribe a deviation $d_t := \text{dist}(r_t, Y_{t-1})$. Thus the entire stepwise realization problem reduces to the scalar distance equation

$$\text{dist}(r_{t-1} - \lambda_t u_t, Y_{t-1}) = d_t. \quad (7)$$

The next result completely characterizes when this equation is feasible and gives the corresponding coefficient λ_t explicitly, thereby turning prescribed approximation geometry into a constructive finite-horizon realization mechanism.

Theorem 2.6 (Constructive controlled-deviation principle). *Under the preceding notation, for each t , define*

$$a_t := (I - \Pi_{Y_{t-1}})r_{t-1}, \quad \alpha_t := \langle a_t, u_t \rangle, \quad d_{t,\min} := \sqrt{\|a_t\|^2 - \alpha_t^2}.$$

Then the prescribed deviation equation (7) is feasible if and only if $d_t \geq d_{t,\min}$, in which case

$$\lambda_t = \alpha_t \pm \sqrt{d_t^2 - d_{t,\min}^2}.$$

Consequently, every feasible deviation profile $(d_t)_{t=1}^T$ can be realized constructively by the updates (6) satisfying exactly $\text{dist}(r_t, Y_{t-1}) = d_t$, for all $t = 1, \dots, T$.

We next instantiate this abstract deviation-control principle on the orbit dictionary used by our realization framework.

Dictionary-induced CCDS realization. To instantiate Theorem 2.6 on the orbit dictionary, let $\mathcal{D} \subset H$ be unit norm. At step t , given Y_{t-1} and r_{t-1} , select an atom $q_t \in \mathcal{D} \setminus Y_{t-1}$ and define

$$\eta_t := \|q_t - \Pi_{Y_{t-1}} q_t\|, \quad u_t := \frac{q_t - \Pi_{Y_{t-1}} q_t}{\eta_t}, \quad Y_t := Y_{t-1} \oplus \text{span}\{u_t\}. \quad (8)$$

For any feasible prescribed deviation $d_t \geq d_{t,\min}$, Theorem 2.6 determines λ_t explicitly and updates

$$\hat{y}_t = \hat{y}_{t-1} + \lambda_t u_t, \quad r_t = y - \hat{y}_t.$$

Since $q_t \notin Y_{t-1}$ and $Y_t = \text{span}\{q_1, \dots, q_t\}$, the selected atoms $q_1, \dots, q_T$ are linearly independent. Moreover, $u_t \in \text{span}\{q_1, \dots, q_t\}$ for every t . Consequently, for every CCDS path $q_{1:T}$ there exists a unique $\beta^{(T)} \in \mathbb{R}^T$ such that

$$\hat{y}_T = \sum_{t=1}^T \lambda_t u_t = \sum_{j=1}^T \beta_j^{(T)} q_j.$$

We call $\beta^{(T)}$ the *CCDS dictionary-coordinate vector*. Its explicit triangular recursion in terms of the innovation coordinates is given in Appendix A.3. We denote this path-specific realization by

$$\text{CCDS}_{\mathcal{D}}^T(y; q_{1:T}, d_{1:T}) := \hat{y}_T,$$

for the CCDS output after T steps. Here $q_{1:T} := (q_1, \dots, q_T)$ denotes the prescribed atom ordering and $d_{1:T} := (d_1, \dots, d_T)$ the prescribed deviation profile. We retain the shorthand $\text{CCDS}_{\mathcal{D}}(y)$ when the ordering and deviation profile are understood.

The nested orthogonal-innovation construction is designed so that, at the minimal-deviation boundary, CCDS recovers the Fourier coefficients of the target in the induced orthonormal basis. Proof of Proposition 2.7 in Appendix A.3.

Proposition 2.7 (Minimal-deviation CCDS recovers the orthogonal coordinates). *Let $\mathcal{D} \subset H$ be a unit-norm dictionary, let $q_1, \dots, q_s \in \mathcal{D}$ be linearly independent, and set $E := \text{span}\{q_1, \dots, q_s\}$. For $y \in H$, apply CCDS with atom ordering $q_{1:s}$ and $d_t = d_{t,\min}$ for $t = 1, \dots, s$. Then*

$$\text{CCDS}_{\mathcal{D}}^s(y; q_{1:s}, d_{\min,1:s}) = \Pi_E y = \sum_{t=1}^s \lambda_t u_t = \sum_{j=1}^s \beta_j^{(s)} q_j,$$

where $\lambda_t = \langle y, u_t \rangle$ and $\beta^{(s)}$ is the CCDS dictionary-coordinate vector.

MP and OMP as minimal-deviation boundaries. Neither Matching Pursuit (MP) nor OMP is, in general, a special case of the other. CCDS, however, provides a common geometric mechanism containing both as boundary regimes. The following proposition identifies MP and OMP as the two minimal-deviation boundary regimes of CCDS while strict interior CCDS updates are globally inaccessible to OMP, marking a geometric separation between OMP and CCDS. Proof of Proposition 2.8 in Appendix A.3.

Proposition 2.8. *Let $\mathcal{D} \subset H$ be a unit-norm dictionary.*

- (i) (MP boundary). *If $Y_{t-1} = \{0\}$ at every step, q_t is selected by the MP atom-selection rule, and $d_t = d_{t,\min}$, then CCDS coincides with MP.*
- (ii) (OMP boundary). *If the selected atoms generate the nested spaces $Y_t = \text{span}\{q_1, \dots, q_t\}$, q_t is selected by the OMP atom-selection rule, and $d_t = d_{t,\min}$, then CCDS coincides with OMP until termination.*
- (iii) (Global geometric separation from OMP). *Any interior CCDS update with $d_t > d_{t,\min}$, $\lambda_t \neq 0$, departing from an OMP boundary state $\hat{y}_{t-1} = \Pi_{Y_{t-1}} y$ is globally inaccessible to OMP, irrespective of its atom selection or ordering.*

Beyond the MP/OMP boundary. The additional freedom of CCDS appears when $d_t > d_{t,\min}$. In this regime, the update is no longer forced to be the extremal orthogonal projection. Instead, CCDS can prescribe any feasible deviation profile $(d_t)_{t=1}^T$ and thereby control the realization path through the nested spaces Y_t .

The strict regime $d_t > d_{t,\min}$ also gives CCDS a pathwise stability property. The perturbation-stability Theorem 2.10 below shows that, when the prescribed CCDS path has a positive feasibility margin, the same atom labels, deviations, and root branches remain feasible under sufficiently small perturbations of the orbit dictionary, and the transported realization varies locally Lipschitz continuously. The boundary case $d_t = d_{t,\min}$, which recovers OMP for the same atom-selection rule, is not covered by this argument because the feasibility margin vanishes. Thus the distinction from OMP is not in endpoint realizability, but the ability to prescribe a realization path and obtain a stability guarantee for that prescribed path.

At the coefficient level, interior choices $d_t > d_{t,\min}$ move CCDS away from the boundary where the root term vanishes in $\lambda_t = \alpha_t \pm \sqrt{d_t^2 - d_{t,\min}^2}$. This avoids boundary-root degeneracy, and when atom selection depends on the evolving residual, can reduce repeated selection of nearly collinear directions; this additional robustness is illustrated in Section 4.

Intrinsic orbit-certificate sparsity. The deviation-control mechanism is compatible with the intrinsic sparse geometry of the orbit dictionary. Let $\mathcal{D}$ be the orbit dictionary, $U = \text{span } \mathcal{D}$, and

$$\mathcal{I} := \{(k, j, m) : 1 \leq k \leq K, 1 \leq j \leq s_k, 0 \leq m \leq d - 1\}.$$

Write $q_{k,j,m} \in \mathcal{D}$ for the corresponding unit-norm orbit atom. For $S \subseteq \mathcal{I}$, define

$$E_S := \text{span}\{q_{k,j,m} : (k, j, m) \in S\}.$$

For $(x_i, y_i) \in \Xi_{\text{train}}$ and $0 \leq \eta \leq \|\Pi_U y_i\|$, set

$$s_i(\eta) := \min \left\{ |S| : S \subseteq \mathcal{I}, \text{dist}(\Pi_U y_i, E_S) \leq \eta \right\}.$$

Thus $s_i(\eta)$ measures the intrinsic per-sample number of orbit-dictionary atoms needed to realize the projected target to tolerance η . Let $S_i = \{r_1, \dots, r_s\}$ be a minimizer, so that $s = |S_i| = s_i(\eta)$. The atoms $q_{r_1}, \dots, q_{r_s}$ are linearly independent. Indeed, if one belonged to the span of the others, removing its index would leave E_{S_i} unchanged, contradicting the minimality of S_i . Proposition 2.7 therefore applies directly to this atom family. Since $E_{S_i} \subseteq U$, we have $\Pi_{E_{S_i}} y_i = \Pi_{E_{S_i}} \Pi_U y_i$. Hence, by the defining property of the minimizing support S_i ,

$$\|\Pi_U y_i - \Pi_{E_{S_i}} y_i\| = \text{dist}(\Pi_U y_i, E_{S_i}) \leq \eta.$$

Thus, we have the following.

Corollary 2.9 (Intrinsic local orbit-certificate sparsity of full CCDS). *With the notation above,*

$$\text{CCDS}_{\mathcal{D}}^s(y_i; q_{r_1:r_s}, d_{\min,1:s}) = \Pi_{E_{S_i}} y_i = \sum_{j=1}^s \beta_j^{(s)} q_{r_j},$$

and

$$\|\Pi_U y_i - \text{CCDS}_{\mathcal{D}}^s(y_i; q_{r_1:r_s}, d_{\min,1:s})\| \leq \eta.$$

In particular, this CCDS realization is supported on at most $s_i(\eta)$ orbit-dictionary atoms.

Repairing CCDS infeasibility. Prescribing the realization path introduces a feasibility obstruction with no direct analogue in standard OMP: for a chosen innovation direction u_t , a desired deviation $d_t < d_{t,\min}$ is infeasible in the current ambient space. This is the finite-dimensional analogue of the prescribed-approximation obstruction underlying Bernstein-type lethargy theory. Such infeasibility can be repaired constructively: by adjoining one orthogonal dimension, any prescribed $0 \leq d_t \leq \|a_t\|$ can be made feasible through a suitable lifted innovation direction. The explicit lifting theorem and its proof are given in Appendix A.3.

Relation between scaffold selection and CCDS realization mechanism. The scaffold selection mechanism is the subspace-level counterpart of the CCDS philosophy introduced later. In scaffold selection, the controlled object is the projection-error functional, and the admissible moves are nested one-dimensional scaffold expansions within the orbit geometry induced by the data. In CCDS realization, the controlled object is the parametrization residual, and the admissible moves are coefficient updates along innovation directions. Thus the two mechanisms are parallel deviation-control theories for the two main bottlenecks of the framework: scaffold geometry and constructive realization. We present scaffold selection first because the framework is logically geometric before constructive: one first selects the support on which the target can be projected, then constructs a realizer on that support.

2.4 Stability under relaxed block interpolation

Exact interpolation provides the canonical construction, with block maps $J_k = Y_k Z_k^\dagger$ recovering the anchor outputs exactly. In practice, however, we use the regularized interpolant $J_{k,\lambda} := Y_k (Z_k^\top Z_k + \lambda I)^{-1} Z_k^\top$, $\lambda > 0$, which is better conditioned on noisy or nearly singular blocks. The next theorem shows that this relaxation preserves the scaffold geometry and perturbs the CCDS realization error only continuously, thereby connecting the exact theory to the implementation used in practice. Proof of Theorem 2.10 in Appendix A.4.

Theorem 2.10 (Stability of the scaffold/CCDS bottlenecks under relaxed interpolation). *Let $A : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be linear, and fix orbit data $(W, \Xi_{\text{sub}}, \{D_k\}_{k=1}^K)$. For each block write Z_k for the orbit matrix and Y_k for the corresponding output matrix as defined in (2), let $J_k = Y_k Z_k^\dagger$ be the reference minimal-norm exact interpolant. Let G be the associated atom matrix and set $U := \text{range}(G)$, $r := \text{rank}(G)$, $\gamma := \sigma_r(G) > 0$. Assume the reference CCDS realizer f_{CCDS} on U has a finite prescribed path with positive feasibility margin and selected raw directions uniformly bounded away from zero. For every admissible relaxed block family $J'_k = Y_k R'_k$, where R'_k is a regularized pseudoinverse factor for the same block data, G' the corresponding atom matrix and $U' := \text{range}(G')$, there exist $\delta_0 > 0$ such that if $\|G' - G\|_2 < \delta_0$, then $U' = U$. Consequently,*

$$\varepsilon_{\text{proj}}(A, U') = \varepsilon_{\text{proj}}(A, U).$$

Moreover, transporting the reference CCDS path to the perturbed atoms, with the same prescribed deviations and the same root branches, gives a CCDS realizer f'_{CCDS} on U' satisfying

$$\varepsilon_{\text{CCDS}}(A, U', f'_{\text{CCDS}}) \leq \varepsilon_{\text{CCDS}}(A, U, f_{\text{CCDS}}) + C_{\text{param}} \|G' - G\|_2.$$

fix notation f_{ccds} , C_{param} Thus, the projection bottleneck is invariant under relaxed interpolation and the constructive CCDS bottleneck is locally Lipschitz stable.

2.5 Our realizability framework in practice.

The preceding theory isolates the projection and parametrization bottlenecks and provides constructive mechanisms for controlling them. In practice, however, scaffold construction begins one step earlier: from the anchor subset Ξ_{sub} , one must choose a partition, orbit depths, and a wrapped shift W that generate a suitable candidate orbit geometry. We now derive observable quantities that guide this preliminary selection and then summarize how they enter the two realizability phases.

Local stability of blockwise interpolation. Equation (3) shows that the blockwise interpolant is sensitive to the conditioning of the orbit matrix. The next result makes this dependence geometric: good conditioning of the input and orbit matrices, together with small displacement between the original anchors and their selected orbit representatives, forces the exact interpolant to remain close to an observable local chart.

Theorem 2.11 (Blockwise exact interpolant as an observable local chart). *Let $D = \{(x_i, y_i)\}_{i=1}^s \subset \mathbb{R}^d \times \mathbb{R}^C$, choose depths $L_1, \dots, L_s$, and define $z_i := W^{L_i} x_i$, $i = 1, \dots, s$, assuming that $z_1, \dots, z_s$ are linearly independent. Set $X := [x_1 \cdots x_s]$, $Z := [z_1 \cdots z_s]$, $Y := [y_1 \cdots y_s]$, and define $J_D := YZ^\dagger$. Let $E := \text{span}\{z_1, \dots, z_s\}$, and let Π_E denote the orthogonal projection onto E . If*

$$\sum_{i=1}^s c_i x_i = 0 \implies \sum_{i=1}^s c_i y_i = 0, \quad (9)$$

then there exists $A_D \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^C)$ satisfying $A_D x_i = y_i$, $i = 1, \dots, s$. Moreover,

$$\|J_D - A_D \Pi_E\| \leq \frac{\|Y\|_F}{\sigma_{\min}^+(X) \sigma_{\min}(Z)} \left(\sum_{i=1}^s \|x_i - W^{L_i} x_i\|^2 \right)^{1/2}, \quad (10)$$

where $\sigma_{\min}^+(X)$ denotes the smallest nonzero singular value of X .

Proof of Theorem 2.11 is deferred to Appendix A.1. Thus well-conditioned blocks with small orbit displacement produce interpolants that remain close to the local geometry of the original anchors.

An observable scaffold diagnostic. The local estimate above can be converted into an anchor-level bound on the projection error of the resulting orbit scaffold. The auxiliary estimate establishing this connection is given in Theorem A.9 in Appendix A.4. Combining it with Theorem 2.11 yields the following observable criterion.

Corollary 2.12 (Observable upper bound for anchor projection error). *Let $P = \{D_1, \dots, D_K\}$ be a partition of Ξ_{sub} , with $D_k = \{(x_{k,1}, y_{k,1}), \dots, (x_{k,s_k}, y_{k,s_k})\}$, and define $X_k := [x_{k,1} \cdots x_{k,s_k}]$, $Y_k := [y_{k,1} \cdots y_{k,s_k}]$, $z_{k,j} := W^{L_{k,j}} x_{k,j}$, $Z_k := [z_{k,1} \cdots z_{k,s_k}]$, $E_k := \text{span}\{z_{k,1}, \dots, z_{k,s_k}\}$. Let U_P denote the orbit scaffold generated by the partition P . Then $\left(\sum_{k=1}^K \sum_{j=1}^{s_k} \|(I - \Pi_{U_P}) y_{k,j}\| \right) / \sum_{k=1}^K s_k \leq \mathcal{S}(P)$, where*

$$\mathcal{S}(P) := \frac{\sum_{k=1}^K \sum_{j=1}^{s_k} \frac{\|Y_k\|_F}{\sigma_{\min}^+(X_k)} \left(\text{dist}(x_{k,j}, E_k) + \frac{\|x_{k,j}\|}{\sigma_{\min}(Z_k)} \left(\sum_{\ell=1}^{s_k} \|x_{k,\ell} - z_{k,\ell}\|^2 \right)^{1/2} \right)}{\sum_{k=1}^K s_k}. \quad (11)$$

The quantity $\mathcal{S}(P)$ therefore provides a fully observable pre-selection criterion for the block structure before the deviation-controlled scaffold-selection procedure is applied. Small $\mathcal{S}(P)$ favors partitions whose local orbit charts are well conditioned and geometrically aligned with the anchors. The number of blocks K provides a complementary complexity criterion: smaller blocks may improve local stability but increase the number of local interpolants J_k that must be learned.

Vanishing-cycle criterion. A second, independent criterion concerns the repeated orbit dynamics of W . Set $\rho := \prod_{i=0}^{d-1} |w_i|$. If $\rho < 1$, then repeated full cycles decay geometrically. Indeed, if $m = qd + r$ with $0 \leq r < d$, Lemma 1.3 gives $W^m = \rho^q W^r$ and hence

$$\|W^m z\| \leq \rho^{\lfloor m/d \rfloor} C_z, \quad C_z := \max_{0 \leq r \leq d-1} \|W^r z\|. \quad (12)$$

Thus, if an input perturbation η is propagated through the orbit, $|W^m \eta| \leq \rho^{\lfloor m/d \rfloor} C_\eta$; hence $\rho < 1$ acts as an implicit noise regularizer by geometrically damping perturbations propagated through repeated orbit cycles, without changing the formal scaffold generated by the d residue-class depths. This is the main reason we favor vanishing shifts in scaffold discovery.

Realizability phases. The practical realization pipeline follows directly from the two-term decomposition. Phase 1 addresses the projection bottleneck by constructing and selecting an orbit scaffold, while Phase 2 addresses the parametrization bottleneck by realizing the projected target on the selected scaffold. Let $\Xi_{\text{sub}} \subseteq \Xi_{\text{train}} \subseteq \Xi$, where Ξ_{sub} is the anchor subset used to construct the orbit scaffold.

Phase 1: scaffold discovery and selection. Phase 1 searches over partitions of the anchor subset Ξ_{sub} , orbit depths, and wrapped shifts. Candidate geometries are guided by three quantities: $\mathcal{S}(P)$, K , $\rho = \prod_{i=0}^{d-1} |w_i|$. Small $\mathcal{S}(P)$ favors good local-chart alignment, moderate K controls the number of local interpolants, and $\rho < 1$ enforces vanishing repeated orbit cycles. The deviation-controlled scaffold-selection procedure of Subsection 2.2 is then applied to the retained candidates to reduce the projection error $\varepsilon_{\text{proj}}$.

Phase 2: CCDS realization. Once the scaffold has been selected, Phase 2 addresses the parametrization bottleneck by realizing each projected training target using the fixed orbit dictionary $\mathcal{D}$ defined in (4). For $(x, y) \in \Xi_{\text{train}}$, we write $\text{CCDS}_{\mathcal{D}}(y) \in U$ for a CCDS realization of $\Pi_U y$. Thus

$$\text{CCDS}_{\mathcal{D}}(y) = \sum_{k=1}^K \sum_{j=1}^{s_k} \sum_{m=0}^{d-1} \beta_{k,j,m}^{\text{CCDS}}(y) J_k W^m x_{k,j}, \quad (x, y) \in \Xi_{\text{train}}. \quad (13)$$

The coefficients $\beta^{\text{CCDS}}(y)$ are produced by the CCDS realization mechanism; CCDS sparsity controls how many orbit-dictionary atoms are active in each realization, as established in Subsection 2.3.

Section 2 has now identified a scaffold, shown that its projected target is constructively and sparsely realizable, and established that this mechanism survives the relaxed interpolation used in practice. Section 3 asks how to extend the resulting sample-wise coefficient rule to unseen inputs.

3 Prediction as Extension of Constructive Orbit-Based Realizability

The motivation for prediction is that Phase 2 already provides, for each training target y_i , a constructive realization $\text{CCDS}_{\mathcal{D}}(y_i)$ of its projection onto the data-induced scaffold. Thus the sample-wise CCDS coefficient rule encodes how the training input–target relation is realized in the fixed orbit geometry. If unseen inputs remain locally close to the training data, it is therefore natural to predict by extending this coefficient rule rather than by learning outputs independently.

Phase 3, the predictive stage of the framework, adds one new object to this construction: a data-dependent coefficient law $\alpha(x) = (\alpha_{k,j,m}(x))_{k,j,m}$, trained on Ξ_{train} to track the effective CCDS coefficients and extend this rule to unseen inputs. The resulting predictor is

$$\tilde{f}(x) := \sum_{k=1}^K \sum_{j=1}^{s_k} \sum_{m=0}^{d-1} \alpha_{k,j,m}(x) J_k W^m x_{k,j}, \quad x \in \mathbb{R}^d. \quad (14)$$

Phase 3: We parameterize the coefficient law α in (14) by a shallow MLP $\alpha_{\theta} : \mathbb{R}^d \rightarrow \mathbb{R}^{|\mathcal{I}|}$. Thus the learned network predicts the coefficients of the fixed orbit representation, rather than the output freely; the final prediction is obtained by the synthesis in (14). The map α_{θ} is trained on the full dataset Ξ_{train} using the effective CCDS coefficients produced in Phase 2 as constructive supervision.

CCDS-supervised Phase-3 training. We train α_{θ} using $L_{\text{Phase 3}} = L_{\text{pred}} + \lambda_{\text{coeff}} \frac{1}{n} \sum_i \|\alpha_{\theta}(x_i) - a^{\text{eff}}(x_i)\|_2 + \lambda_{\text{CCDS}} \frac{1}{n} \sum_i \|\tilde{f}_{\theta}(x_i) - \text{CCDS}_{\mathcal{D}}(y_i)\|_2^2 + \lambda_1 \frac{1}{n} \sum_i \|\alpha_{\theta}(x_i)\|_1$, where any of the regularization weights may be set to zero.

Definition 3.1 (Weighted Backward Shift Neural Network). *A Weighted Backward Shift Neural Network (WBSNN) associated with Ξ_{train} is the predictor $\tilde{f}$ of (14) produced by the three-phase construction of this paper.*

Structural interpretability. WBSNN is structurally interpretable at the realization level: Phase 1 exposes the data-induced scaffold, Phase 2 constructs the corresponding CCDS realizations, and Phase 3 expresses every prediction through the coefficient vector $\alpha_{\theta}(x)$ in the fixed orbit representation. Consequently, every prediction admits an explicit coefficient decomposition over the fixed orbit representation, making the contribution of the learned orbit coordinates directly inspectable. The MLP producing α_{θ} is not itself claimed to be fully transparent.

Prediction guarantees. Throughout the prediction guarantees below, $\mathcal{D}$ and $U = \text{span } \mathcal{D}$ denote the orbit dictionary and scaffold defined in (4), and $\text{CCDS}_{\mathcal{D}}$ denotes the CCDS realization map defined above. On Ξ_{train} , let $f(x_i) := y_i$, so that $\text{CCDS}_{\mathcal{D}} \circ f$ denotes the map $x_i \mapsto \text{CCDS}_{\mathcal{D}}(y_i)$, the Phase 2 realization of $\Pi_U y_i$. The Phase 3 predictor $\tilde{f}$ is the coefficient-law predictor defined in (14).

Let $\Xi_{\text{train}} = \{(x_i, y_i)\}_{i=1}^n = D_1 \sqcup \dots \sqcup D_K \subset \mathbb{R}^d \times \mathbb{R}^C$, $\Xi_{\text{test}} = \{(\bar{x}_j, \bar{y}_j)\}_{j=1}^m \subset \mathbb{R}^d \times \mathbb{R}^C$. Define the dataset seminorm $\|g\|_{\Xi} := (\sum_{x:(x,y) \in \Xi} \|g(x)\|_{\ell^2})/|\Xi|$ for maps g defined on $\{x : (x, y) \in \Xi\}$.

The first result shows that prediction on the training set inherits the two realizability bottlenecks and adds only the Phase 3 approximation error.

Theorem 3.2 (Geometric in-sample prediction bound). *Assume that, for each $i = 1, \dots, n$, $\|f(x_i) - \text{CCDS}_{\mathcal{D}}(y_i)\|_{\ell^2} \leq \delta_i$. Then*

$$\|f - \tilde{f}\|_{\Xi_{\text{train}}} \leq \underbrace{\|f - \Pi_U f\|_{\Xi_{\text{train}}}}_{\varepsilon_{\text{proj}}(f, U; \Xi_{\text{train}})} + \underbrace{\|\Pi_U f - \text{CCDS}_{\mathcal{D}} \circ f\|_{\Xi_{\text{train}}}}_{\varepsilon_{\text{param}}(f, U, \text{CCDS}_{\mathcal{D}} \circ f; \Xi_{\text{train}})} + \bar{\delta}, \quad (15)$$

where $\bar{\delta} := \frac{1}{n} \sum_{i=1}^n \delta_i$.

The proofs of both Theorems 3.2 and 3.3, which follow from elementary triangle-inequality arguments, are deferred to the Appendix A.5.

To pass from in-sample realization to generalization, we additionally require test points to remain close to the geometry observed at training. Under this assumption, the next result extends the same structural decomposition out of sample, with an additional geometric transfer term.

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be the target map, so that $f(x_i) = y_i$, $f(\bar{x}_j) = \bar{y}_j$. Let $\tilde{f}$ be the predictor built on the scaffold U .

Theorem 3.3 (Geometric out-of-sample prediction bound). *Assume*

1. (Neighborhood relation between test and train) *There exists $\delta \geq 0$ such that for every $(\bar{x}_j, \bar{y}_j) \in \Xi_{\text{test}}$ there exists some $(x_{i(j)}, y_{i(j)}) \in \Xi_{\text{train}}$ with $\|\bar{x}_j - x_{i(j)}\|_{\ell^2} \leq \delta$.*
2. (Local regularity of the target) *There exists $L_f \geq 0$ such that, for every matched pair $\bar{x}_j, x_{i(j)}$ from Assumption 1, $\|f(\bar{x}_j) - f(x_{i(j)})\|_{\ell^2} \leq L_f \|\bar{x}_j - x_{i(j)}\|_{\ell^2}$.*
3. (Regularity of the predictor) *The map $\tilde{f}$ is $L_{\tilde{f}}$ -Lipschitz on $\Xi_{\text{train}} \cup \Xi_{\text{test}}$: $\|\tilde{f}(x) - \tilde{f}(x')\|_{\ell^2} \leq L_{\tilde{f}} \|x - x'\|_{\ell^2}$, $\forall (x, y), (x', y') \in \Xi_{\text{train}} \cup \Xi_{\text{test}}$.*

Then

$$\begin{aligned} \|f - \tilde{f}\|_{\Xi_{\text{test}}} &\leq \underbrace{\|f - \Pi_U f\|_{\Xi_{\text{test}}}}_{\varepsilon_{\text{proj}}(f, U; \Xi_{\text{test}})} + \underbrace{\max_{x:(x,y) \in \Xi_{\text{train}}} \|\Pi_U f(x) - \text{CCDS}_{\mathcal{D}}(f(x))\|_{\ell^2}}_{\text{CCDS realizability error}} \\ &\quad + \underbrace{\max_{x:(x,y) \in \Xi_{\text{train}}} \|\text{CCDS}_{\mathcal{D}}(f(x)) - \tilde{f}(x)\|_{\ell^2}}_{\text{Phase 3 CCDS-tracking error}} + (L_f + L_{\tilde{f}})\delta. \end{aligned}$$

Remark 3.4 (On the Lipschitzness of the Phase 3 predictor). *Let $S_{\mathcal{D}} : \mathbb{R}^P \rightarrow \mathbb{R}^C$ denote the fixed synthesis check notation scaffold map $S_{\mathcal{D}}(a) = \sum_{p=1}^P a_p d_p$. Since $\tilde{f}_{\theta} = S_{\mathcal{D}} \circ \alpha_{\theta}$, if α_{θ} is L_{α} -Lipschitz, then $\tilde{f}_{\theta}$ is Lipschitz with $L_{\tilde{f}} \leq \|S_{\mathcal{D}}\|_{L_{\alpha}}$. In particular, this holds for the shallow MLP used in Phase 3 whenever its constituent layers and activations are Lipschitz.*

merge all notations at the beginning xi train, sub, test etc

Table 1 summarizes the geometric and realizability diagnostics of Phases 1 and 2, while Table 2 summarizes the predictive coefficient diagnostics of Phase 3. Across all representative datasets, the same qualitative

Table 1: Interpretability diagnostics for Phases 1 and 2 of main experiments for WBSNN, subsection 4.2. Lower projection residual and lower interpolation/parametrization errors indicate a scaffold that captures the target geometry well and supports accurate constructive realization. Values are mean $\pm$ standard deviation over seeds.

Dataset	$\varepsilon_{\text{proj}}$	$\varepsilon_{\text{param}}$
MNIST	$5.32\text{e-}15 \pm 2.2\text{e-}15$	$1.46\text{e-}7 \pm 6.21\text{e-}9$
ISOLET	$3.11\text{e-}9 \pm 6.33\text{e-}10$	$3.25\text{e-}6 \pm 2.47\text{e-}8$
Gas Sensor Drift	$4.21\text{e-}11 \pm 6.03\text{e-}11$	$8.92\text{e-}8 \pm 3.89\text{e-}8$

Table 2: Interpretability diagnostics for Phase 3 of main experiments for WBSNN, subsection 4.2. Lower coefficient-tracking error and smaller effective support indicate closer adherence to the constructive CCDS rule. Higher top-3 mass and higher neighbor support overlap indicate stronger coefficient concentration and greater local support stability. Values are mean $\pm$ standard deviation over seeds.

Dataset	E_{coeff}	Support size	Top-3 mass	Neighbor overlap
MNIST	$1.47\text{e-}1 \pm 6.22\text{e-}3$	18.53 ± 0.23	$(93.13 \pm 0.5)\%$	0.663 ± 0.016
ISOLET	$3.26\text{e-}1 \pm 3.35\text{e-}3$	24.92 ± 0.16	$(35.2 \pm 0.4)\%$	0.705 ± 0.015
Gas Sensor Drift	$7.05\text{e-}2 \pm 1.47\text{e-}2$	9.73 ± 0.64	$(97.48 \pm 0.4)\%$	0.85 ± 0.028

pattern is observed: the selected scaffold captures a large portion of the target geometry, the Phase 2 realizer fits that projected target with low residual, and the final hybrid predictive phase remains sparse, concentrated, and close to the effective CCDS rule. This supports the intended interpretation of WBSNN as a structured orbit-based pipeline rather than as a black-box predictor.

4 Experiments

We instantiate the constructive theory in two experimental regimes. First, we study realizability, where the goal is to test whether a target operator can be constructively synthesized on a selected orbit scaffold, and to isolate the roles of projection quality and deviation-controlled realization. Second, we study predictability, where the same scaffold-realization mechanism is used inside WBSNN, with Phase 3 implemented by a learned coefficient law over the fixed orbit-derived representation, supervised by the constructive Phase 2 CCDS realizations, rather than by an unconstrained output classifier. Throughout, the experiments are designed to test the theory’s structural claims, not merely to maximize benchmark scores.

Unless otherwise stated, all features are standardized using training-set statistics only and compressed by PCA before Phase 1. This imposes a controlled low-dimensional information bottleneck in which scaffold geometry, constructive realizability, and predictive coefficient tracking can be examined consistently. In the predictive experiments, WBSNN and all matched baselines operate on the same PCA-compressed features; raw-input or domain-specific models, when reported, are identified separately as reference comparisons. Only a small *discovery fraction* of the training set is used to build the orbit scaffold. All experiments run in a lightweight CPU-only Jupyter environment—no GPUs—so the numbers reflect true data/geometry leverage rather than horsepower. The experiments are therefore not designed around a large-scale engineering scaling narrative, but around illustrating the practical consequences of the theory in controlled finite-dimensional regimes through scaffold quality, constructive realizability, and predictive coefficient tracking. Across all experiments, both the CCDS realizer (Phase 2) and the learned coefficient law (Phase 3) exhibit strong sparsity (see diagnostic tables).

Phase 1/2 protocol. Phase 1 uses the cheaper and more stable regularized pseudoinverse interpolant: this choice is fully justified by Theorem 2.10 and Remark ???. Phase 2 uses CCDS to realize the projected target on the selected scaffold. We report two implementations. The fast greedy MP/CCDS solver selects atoms by the matching-pursuit correlation rule, but updates coefficients through the MP/CCDS distance equation; hence it is MP only in atom selection, not in its full update rule. The full CCDS solver instead uses the nested-subspace innovation construction and is reported separately to show how prescribed deviation control affects convergence under the same selected scaffold.

4.1 Experiments on realizability

The realizability experiments isolate the two bottlenecks from Section 2: the projection bottleneck, governed by scaffold quality, and the parametrization bottleneck, governed by constructive synthesis on the selected scaffold. The first two experiments already expose the necessity of CCDS under weak or degenerate geometry. We then add one diagnostic experiment whose sole purpose is to make scaffold quality visible as an empirical object.

Experiment 1: Pathological weak-alignment stress test and deviation-control ablation. We evaluate Phase 2 realizability on an extreme synthetic weak-alignment dataset designed to destroy global dictionary coherence. Samples are generated from random label-dependent directions in independent orthogonal frames and are then re-rotated by independent random rotations; large label noise further places residual directions in near-generic position relative to any fixed orbit dictionary. This creates a small-cosine regime in which greedy one-ray alignment is unreliable and CCDS updates are expected to dominate.

We use an ill-conditioned operator family with no PCA, working in full dimension $d = 10$ (raw scaled features), 500 training samples, seven output classes, and a strict Phase 1 budget of 7% of the training set. The selected scaffold uses only 35 anchor samples, organized into 23 blocks, and reduces the projection bottleneck to numerical precision, $\varepsilon_{\text{proj}} \approx 1.35 \times 10^{-10}$. Thus the experiment no longer tests failure caused by scaffold incompleteness; instead, it isolates the ability of MP/CCDS to realize projected targets under weak alignment. The empirical alignment diagnostic confirms the intended pathology: 0% of probe samples maintain $\cos \theta \geq 0.8$ over the first 20 greedy steps, and the MP certification rate is 0%. The fast MP/CCDS solver nevertheless realizes the projected operator outputs to numerical precision, with $\varepsilon_{\text{param}} \approx 6.36 \times 10^{-8}$, using a mean of 14.11 steps. Hence the experiment shows that weak one-step alignment does not prevent constructive realization once the scaffold has captured the relevant output geometry. Full CCDS, run as a deviation-control diagnostic, reveals the expected dependence on the prescribed contraction schedule. For $\gamma_{\text{dev}} = 0$, where d_t is driven to the minimal feasible boundary, full CCDS obtains $\varepsilon_{\text{param}} \approx 2.2 \times 10^{-7}$ and a 99.2% realizability rate (i.e. 99.2% of training samples have $\varepsilon_{\text{param}} \leq 10^{-6}$). For $\gamma_{\text{dev}} = 0.5$, the imposed interior deviation profile is more conservative, increasing the mean residual to approximately 5.33×10^{-4} and reducing exact realizability to 54.4%. For $\gamma_{\text{dev}} = 0.9$, the residual contraction is more conservative within the fixed step budget: all samples reach the step limit and the full-CCDS realizability rate drops to 41.2% reaching a mean residual of 1.18×10^{-3} .

This ablation separates three effects. First, scaffold selection can collapse the geometric projection bottleneck even in an adversarial weak-alignment dataset. Second, fast MP/CCDS can still synthesize the projected target to numerical precision, despite the absence of certified pure-MP trajectories. Third, full CCDS should not be interpreted as a residual-minimizing solver in this diagnostic: as γ_{dev} increases, it deliberately realizes an interior deviation profile rather than the smallest attainable residual. The sparse and class-structured atom usage shows that the realizations remain interpretable even in this pathological regime. We report the full ablation in Table 7 and provide representative solver traces in the complementary material.

Experiment 2: Extreme-Ray Dominance vs. Interior Feasibility in Near-Parallel Subspaces Under Extremely Low Budget Random Scaffolds. We evaluate Phase 2 realizability of MP/CCDS on a challenging synthetic dataset formed by a union of six nearly parallel 3-dimensional subspaces in $\mathbb{R}^{35}$ ($\sigma_{\text{parallel}} = 0.02$, $\sigma_{\text{noise}} = 0.05$), reduced to $d = 15$ by PCA (98.2% variance retained) and 2000 training samples. Labels are generated by linear operators C followed by a fixed readout B , yielding classification tasks with identical input geometry but diverse operator spectra (identity, orthogonal, ill-conditioned, low-rank, spiked, sparse, Toeplitz, and adversarially scaffold-misaligned operator). A deliberately weak scaffold coverage (3% random subset, no optimization) is used to decouple Phase 2 realizability from scaffold selection. Notably, the low-rank operator is the only case with a large projection error. This highlights a limitation of rank-based diagnostics often used in machine learning: low intrinsic rank does not necessarily imply geometric compatibility with the representation space on which realization is attempted. A low-dimensional operator can still be poorly captured when its active range directions are misaligned with the dataset-induced scaffold. In this sense, scaffold alignment provides a finer diagnostic than rank alone.

Despite poor projection coverage, fast MP/CCDS consistently achieves extremely small parametrization error ($\varepsilon_{\text{param}} \sim 10^{-8}$ – 10^{-6}) via very sparse representations. However, strict MP contraction is almost never certified: residuals typically lie on flat or symmetric faces of the feasible cone, where several atoms have nearly equal correlations and no single extreme ray has a margin. Full CCDS, by contrast, enforces deviation-controlled multi-atom feasibility rather than one-step contraction, achieves moderate to high realizability (4–88% of training samples have Full CCDS parametrization error below $1\text{e-}6$) across spectra, constructing stable interior two-atom certificates, trading minimal residual for interior feasibility. More details on full diagnostics and interpretability metrics in Tables 3 and 8 in Appendix A.

Furthermore, the fast MP/CCDS and full CCDS traces for Sample 0 (Identity and Orthogonal operator), illustrate the sharp contrast between greedy dynamics and deviation-controlled certificates (full details in complementary materials). In fast mode, the Identity operator case converges in 14 steps with occasional CCDS fallbacks, while the Orthogonal operator case requires 50 steps and exhibits strong alternation between a few atoms (e.g., repeated switching between 327 and 194), a clear instance of boundary crawling along a flat cone face. This directly visualizes the margin-free pathology: no single atom dominates, so pure MP progresses slowly and fragily. In contrast, full CCDS always produces a clean two-step interior-feasible certificate. Together, these traces provide concrete evidence that greedy contraction is insufficient in near-parallel geometries, whereas deviation control yields short, stable, and geometrically robust realizations.

Overall, Experiment 2 exposes a core limitation of classical greedy contraction: MP requires strict single-ray dominance and fails almost universally due to margin collapse under symmetry, degeneracy, or near-parallelism. Realizability is instead governed by cone interior width and operator-induced anisotropy. This benchmark cleanly separates approximation quality from contraction geometry and shows that deviation-controlled solvers such as CCDS are essential — not optional — for achieving reliable, stable realizability under weak, unoptimized scaffolds. For a full report of interpretability diagnostics (per-sample supports, atom usage histograms, deviation paths, and stopping statistics), we refer the reader to the complementary materials.

Experiment 3: scaffold-quality diagnostic. This experiment is designed to empirically validate the scaffold bottleneck itself. We reuse the same synthetic generator as in Experiment 1 so that the operator family, ambient dimension, and noise model remain fixed; only the scaffold construction mechanism is varied. For each of 10 runs, we construct three candidate scaffolds with the same scaffold-size budget and the same realization-step budget: a selected orbit scaffold produced by the scaffold-selection algorithm from Section 2.2, a random orbit scaffold produced from the same orbit-generated family but with a random admissible subset/partition choice, and a random non-orbit based Gaussian scaffold built from normalized Gaussian atoms with the same total number of atoms as the orbit dictionary. For the Gaussian baseline we impose a proper output-rank bottleneck $r \leq C - 1$, since a full-rank Gaussian scaffold spans the entire output space almost surely and makes $\varepsilon_{\text{proj}}$ degenerate. For each scaffold U , we compute projection error, parametrization error and the *explained label-space energy* defined as $E_{\text{expl}}(U) = \frac{\|\Pi_U Y\|_F^2}{\|Y\|_F^2}$. Table 9 in Appendix A reports the results. The selected orbit scaffold dominates both comparison scaffolds—the random orbit scaffold and the random non-orbit based Gaussian scaffold—not only in projection quality but also in downstream realizability. This experiment is especially important because it shows directly that orbit-based scaffolds do matter for learning: the gain is not coming merely from dictionary size or ambient dimension, but from the orbit structure itself and from selecting that structure in a data-adaptive way. The systematic gap between the selected orbit scaffold, the random orbit scaffold, and the random non-orbit based Gaussian baseline therefore gives concrete empirical evidence that learnability in our framework is driven by scaffold geometry.

4.2 Experiments on predictability

We now evaluate the predictive instantiation of the framework. The predictive section is intentionally compact but sufficient to illustrate the theory: three representative predictive runs and three focused ablations spanning five datasets. The datasets serve distinct roles: chronological distribution shift (Gas Sensor Drift), high-dimensional multiclass speech recognition under feature compression (ISOLET), compressed vision and

Table 3: Experiment 2: Near-parallel and symmetric degeneracy benchmark with random 3% scaffold. Full CCDS remains robust when greedy single-ray contraction becomes unstable or vacuous.

Operator	% samples MP updates	% samples CCDS updates	Fast $\varepsilon_{\text{param}}$ (mean)	Full CCDS $\varepsilon_{\text{param}}$ (mean)	Steps (fast mean)	$\varepsilon_{\text{proj}}$ (mean)
I	0.20	99.80	2.06e−06	3.43e−06	20.76	2.56e−06
ORTHO	0.10	99.90	5.09e−07	2.90e−06	20.97	2.60e−06
ILL_SVD	0.00	100	1.31e−07	1.02e−06	21.24	6.87e−07
LOWRANK	0.00	100	1.84e−07	7.26e−07	18.09	7.18e−01
SPIKED	0.00	100	2.20e−07	1.60e−06	20.41	1.44e−06
SPARSE	0.00	100	1.88e−07	1.38e−06	20.63	1.10e−06
TOEPLITZ	0.00	100	5.64e−08	6.40e−07	20.79	2.58e−07
ADV_SCAFF	0.05	99.95	6.73e−07	2.56e−06	20.93	2.40e−06

discovery-budget saturation (MNIST), recovery of hidden manifold geometry (Swiss roll + RFF), and stability of exact versus relaxed interpolation under controlled label noise (noisy synthetic gauntlet).

Baselines. We compare against strong baselines organized by family and matched to the regime under consideration, but we deliberately keep the main-paper suite compact. Because all predictive experiments are conducted in PCA-compressed finite-dimensional settings, we restrict attention to controls that are meaningful under the same compressed-feature protocol: one strong linear baseline, one strong kernel baseline, one strong tree ensemble, one shallow neural baseline, and, when the dataset warrants it, one domain-matched baseline tailored to the underlying shift or drift structure. This preserves the family-organized comparison principle without turning the main text into an exhaustive benchmark inventory.

Main predictive results. The main predictive results are reported in Tables 1, 2 and 4.

Gas Sensor Drift. We use the batch-chronological protocol, training on earlier batches and testing strictly on later batches under sensor drift. Features are standardized on the training split and compressed to $d = 10$ principal components (95% retained variance). We use 3000 training samples. The scaffold is discovered from only a 1% subset of the training data. This experiment therefore evaluates predictive performance under simultaneous distribution shift and an extremely limited discovery budget. Despite relying on a highly sparse orbit realization over the selected scaffold (9.7 atoms in average used per sample out of 220), WBSNN outperforms the considered baselines showing that accurate prediction can be achieved with a compact and interpretable orbit representation rather than dense parameterization.

ISOLET. We evaluate on the ISOLET spoken letter recognition dataset using PCA-compressed features $d = 25$ (71% retained variance). With the full training split of 6238 samples, the scaffold is discovered from only a 1% subset. The fast MP/CCDS realizer proves insufficient, producing a relatively high parametrization error ($\approx 5.2 \times 10^{-3}$) with no samples achieving pure MP certification. We therefore fall back to the full CCDS solver, which attains near machine-precision realization ($\approx 3.26 \times 10^{-6}$) across all samples. The resulting orbit representation is highly sparse: each prediction activates on average only 24.9 atoms out of the nominal 550-atom scaffold, with the top three coefficients carrying approximately 35.2% of the mass. Despite the extremely limited discovery budget, the WBSNN predictor trained on these coefficients reaches 91.94% test accuracy, remaining competitive with strong baselines while relying on a compact and explicitly constructed sparse orbit representation.

MNIST. We use PCA-compressed pixel features with $d = 25$ (46% retained variance), 5000 training samples, and a 1% discovery fraction for scaffold selection. This experiment evaluates whether a small orbit scaffold can support accurate prediction from a highly compressed representation. Despite relying on a sparse orbit realization learned from this limited discovery budget (18.5 atoms in average used per sample out of 525), WBSNN outperforms all classical baselines trained on the same compressed features, including SVM and Gradient Boosting, while remaining fully interpretable through its scaffold and sparse CCDS representation. Convolutional architectures such as CNNs achieve higher accuracy by operating directly on the full training

set of raw images, rather than learning from a 1% scaffold over compressed features. Even under this substantially more favorable setting, the performance gap remains relatively small.

For the full set of interpretability diagnostics introduced in subsection ??, we refer the reader to the complementary materials.

Table 4: Main predictive results under the updated protocol: scaffold selection in Phase 1, CCDS-based realization in Phase 2, and hybrid CCDS coefficient tracking in Phase 3. For each dataset we report strong baselines organized by family and matched to the compressed-feature regime: one representative linear, kernel, tree, and shallow neural baseline, plus a domain-matched baseline when relevant. Results are mean $\pm$ standard deviation over seeds. Best result in bold.

Dataset	WBSNN 1%	LogReg	RBF-SVM	GBT	MLP	Domain-matched
MNIST	0.934 ± 0.003	0.877 ± 0.002	0.930 ± 0.001	0.919 ± 0.002	0.908 ± 0.01	CNN (raw img) 0.971 ± 0.01
ISOLET	0.919 ± 0.006	0.915 ± 0.0	0.929 ± 0.0	0.891 ± 0.004	0.903 ± 0.007	Nystrom RBF 0.912 ± 0.00
Gas Sensor	0.71 ± 0.03	0.61 ± 0.05	0.56 ± 0.02	0.64 ± 0.03	0.67 ± 0.05	LogReg+CORAL 0.61 ± 0.05

Predictive ablations

Ablation 1: recovering hidden manifold geometry through scaffold selection. Does scaffold selection recover the intrinsic manifold geometry better than random orbit or unstructured Gaussian scaffolds? This ablation isolates the contribution of scaffold geometry itself and is among the most important experiments in the paper. We use the noisy 5-class Swiss-roll with random Fourier features dataset precisely because the ground-truth manifold geometry is known while the observed representation is nonlinear, noisy and high-dimensional (30-D RFF features projected to PCA = 10) (58.4% retained variance) and 3000 training samples. All reported values are mean $\pm$ standard deviation across 10 independent runs performed with paired random seeds. This setup lets us test directly whether predictive performance requires recovering the right orbit geometry or whether it can be explained merely by dictionary size or Phase-3 capacity. Under identical Phase-1/2 budgets ($p = 1\%$) and with the same Phase-3 coefficient head, we compare three scaffolds of comparable atom count: (i) the data-adaptive orbit scaffold returned by our selection algorithm, (ii) a random orbit scaffold drawn from the same orbit-generated family, and (iii) an unstructured Gaussian scaffold. Table 5 shows a clear, systematic ordering: the selected orbit scaffold achieves near machine-precision alignment with the target $\varepsilon_{\text{proj}} = 1.37 \times 10^{-10} \pm 3.9 \times 10^{-10}$, while a random orbit scaffold of the same size yields $\varepsilon_{\text{proj}} = 1.96 \times 10^{-2} \pm 3.4 \times 10^{-2}$ and an unstructured Gaussian scaffold yields $\varepsilon_{\text{proj}} = 4.36 \times 10^{-1} \pm 8.2 \times 10^{-2}$. Consistent with the scaffold quality metrics, test accuracy degrades sharply with increasing variance: Selected orbit: 0.8689 ± 0.0021 , Random orbit: 0.8293 ± 0.0725 , Gaussian: 0.7247 ± 0.1068 . This ordering demonstrates that both ingredients are essential. First, orbit structure matters: random orbit scaffolds consistently outperform unstructured Gaussian scaffolds, showing that the orbit-generated family carries genuine predictive content beyond mere dictionary expansion. Second, scaffold selection matters: the data-adaptive selected orbit scaffold consistently outperforms a random orbit scaffold of identical size, proving that learnability is not driven by orbit structure in the abstract but by recovering the specific orbit geometry compatible with the dataset. In short, the ablation supplies direct empirical support for the central claim of the paper: prediction improves when the scaffold captures more of the underlying target action, and this improvement arises from structured, data-adaptive orbit geometry rather than from brute-force growth of dictionary capacity.

Ablation 2: exact vs. relaxed interpolation under controlled label noise. Theorem 2.10 turns exact interpolation from a fragile idealization into a practical principle: the exact interpolant is the canonical reference object, but a regularized relaxed interpolant can be used without destroying the scaffold geometry. Indeed, the theorem predicts that replacing the exact block interpolants by their regularized pseudoinverse versions should preserve the projection bottleneck and perturb the CCDS parametrization bottleneck only continuously, as long as the induced atom system remains in the local stability regime. We test this prediction on a noisy synthetic gauntlet with 15,000 samples and 50 raw features, reduced by PCA to $d = 10$ (94.1% retained variance). The dataset combines informative coordinates, correlated nuisance coordinates, and a mildly spurious feature. We add symmetric noise only to the training labels at levels 0%, 20%, and 40%, while keeping test labels clean. Exact and relaxed interpolation are run with paired seeds and the same 1% Phase

1/2 discovery budget. In the relaxed arm we use a fixed ridge parameter $\lambda = 10^{-3}$, so the near-identical test accuracies are obtained despite using a genuinely regularized, non-exact local interpolant.

The outcome is striking: across all noise levels, the test accuracies of exact and relaxed interpolation remain essentially identical. Thus, machine-precision anchor interpolation is not the fragile ingredient driving prediction. What survives—and what Theorem 2.10 identifies—is the stability of the orbit scaffold and of the CCDS coefficient law induced on it. This justifies using the regularized relaxed interpolant as the main protocol: it keeps the predictive behavior of the exact construction while being cheaper, better conditioned, and more robust under noisy local blocks.

Ablation 3: discovery-budget curve. This ablation varies the Phase 1/2 discovery fraction while keeping all other settings fixed. Its purpose is to test whether the orbit scaffold needed for prediction can be identified from a small discovery set. Results are reported in Table 10 in Appendix A. The main pattern is that predictive performance saturates very early. On MNIST (5000 training samples, PCA = 20, 42% retained variance), moving from 1% to larger discovery budgets changes test accuracy only marginally (a few tenths of a percentage point). On Gas Sensor Drift (3000 training samples, PCA = 10, 95% retained variance), the 1% discovery budget is in fact the best. Across budgets, the projection and parametrization errors remain numerically negligible, so the scaffold and CCDS realization bottlenecks have already been solved and the limiting factor has shifted to Phase 3. Thus, the ablation supports a data-efficient scaffold-discovery principle. Once a small orbit scaffold captures the relevant label-space geometry, adding more discovery anchors mainly enlarges the dictionary and introduces additional coefficient degrees of freedom; under drift, these extra degrees of freedom can even reduce transfer stability. The relevant conclusion is therefore not “more discovery data gives better accuracy,” but rather that WBSNN can recover a predictive orbit scaffold from a very small discovery budget, after which generalization is governed by the transferability of the learned CCDS coefficient law. In this sense, small discovered scaffolds appear to act as structural regularizers, especially in drifted regimes.

Table 5: Ablation 1: Scaffold ablation on Swiss Roll + RFF. The selected orbit scaffold achieves near machine-precision alignment with the target and outperforms both random orbit and Gaussian scaffolds of the same size in test accuracy, while also exhibiting much lower variance.

Scaffold type	Test acc	$\varepsilon_{\text{proj}}$	$\varepsilon_{\text{param}}$	Effect Support
WBSNN; Selected Orbit (1%)	0.8689 ± 0.0021	1.37e−10 ± 3.9e−10	5.32e−8 ± 4.2e−8	5.63 ± 1.44
WBSNN; Random Orbit (1%)	0.8293 ± 0.0725	1.96e−2 ± 3.4e−2	1.55e−8 ± 7.4e−9	3.35 ± 0.85
Gaussian Scaffold (1%)	0.7247 ± 0.1068	4.36e−1 ± 8.2e−2	5.34e−3 ± 8.5e−3	1.41 ± 1.46

Table 6: Ablation 2: Exact vs relaxed interpolation under controlled training-label noise. Test labels are kept clean. Values are mean ± st dev over five seeds. Near-identical test accuracies show that the relaxed regularized interpolant preserves the predictive behavior of the exact scaffold construction, in agreement with Theorem 2.10.

	Test acc (0% noise)	Test acc (20% noise)	Test acc (40% noise)
WBSNN, Phase 1/2: 1 % relaxed	0.9711 ± 0.0015	0.9691 ± 0.0012	0.8925 ± 0.0077
WBSNN, Phase 1/2: 1 % exact	0.9710 ± 0.0010	0.9694 ± 0.0008	0.8925 ± 0.0077
Best baseline	0.975 ± 0.0 DRO-Log	0.958 ± 0.0 LogReg	0.9327 ± 0.0 LogReg
Worst baseline	0.925 ± 0.0 ExtraTrees	0.9163 ± 0.0 ExtraTrees	0.7689 ± 0.001 RdmFor

Conclusion.

This work establishes a unified orbit-based language for scaffold discovery, constructive realization, and prediction through a single local-to-global orbit scaffold-realization principle, laying the foundation for a broader structural theory in which seemingly distinct learning problems may be understood as instances of the same underlying mechanism.

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A Proofs and Remaining Tables

To prove Lemma 1.2, we first establish its action on the canonical basis.

Lemma A.1. *For every $m \geq 1$, $W^m e_j = \left(\prod_{s=0}^{m-1} w_{j-s \pmod{d}} \right) e_{j-m \pmod{d}}$.*

Proof. Indeed, the claim follows by induction. For $r = 1$, it is the definition. Assuming it for r ,

$$W^{r+1} e_j = W \left[\left(\prod_{s=0}^{r-1} w_{j-s} \right) e_{j-r} \right] = \left(\prod_{s=0}^{r-1} w_{j-s} \right) w_{j-r} e_{j-r-1} = \left(\prod_{s=0}^r w_{j-s} \right) e_{j-r-1},$$

where all indices are taken modulo d . □

Proof of Lemma 1.2. Write $z = \sum_{q=0}^{d-1} z_q e_q$. By Lemma A.1,

$$W^m z = \sum_{q=0}^{d-1} z_q \left(\prod_{s=0}^{m-1} w_{q-s \pmod{d}} \right) e_{q-m \pmod{d}}.$$

The coefficient of e_{i-1} is obtained for $q = i + m - 1 \pmod{d}$. Therefore

$$(W^m z)_{i-1} = z_{(i+m-1) \pmod{d}} \prod_{s=0}^{m-1} w_{i+m-1-s \pmod{d}}.$$

Reindexing with $r = m - 1 - s$ gives

$$(W^m z)_{i-1} = \left(\prod_{r=0}^{m-1} w_{(i+r) \pmod{d}} \right) z_{(i+m-1) \pmod{d}}, \quad i = 1, \dots, d.$$

For $m = 0$, this reduces to $W^0 = I$. □

A.1 Exact interpolation Theorem in infinite dimensions

Thanks to Hahn-Banach Theorem, we show that for infinite dimensional Banach spaces, under a technical independence condition, any finite dataset admits an exact realization within a class of transformed weighted shifts. We now present and prove an explicit construction of such a realization.

Theorem A.2 (Exact interpolation along independent orbits). *Let X and Y be infinite-dimensional Banach spaces. Let $T : X \rightarrow X$ be a bounded linear operator and let $\{(x_i, y_i)\}_{i=1}^{n_0} \subset X \times Y$ be a finite input-target set. Suppose that for each $i = 1, \dots, n_0$ there exists a nonnegative integer L_i such that*

$$z_i := T^{L_i} x_i \notin \text{span}\{z_1, \dots, z_{i-1}\}, \quad i \geq 2. \tag{16}$$

Then there exists a bounded linear operator $J : X \rightarrow Y$ such that $JT^{L_i} x_i = y_i$, $i = 1, \dots, n_0$.

Proof. We construct J recursively. Let $J_0 := 0 \in \mathcal{L}(X, Y)$. For $i = 1$, pick any $L_1 \geq 0$, set $z_1 := T^{L_1} x_1$ and find a linear functional $f_1 \in X^*$ with $f_1(z_1) = 1$. Define $J_1 := J_0 + (y_1 - J_0 z_1) \otimes f_1$, where $(y \otimes f)(v) := f(v)y$. Equivalently, for all $v \in X$ we have $J_1 v := J_0 v + (y_1 - J_0 z_1) \otimes f_1(v) = 0 + (y_1 - 0) \otimes f_1(v) = f_1(v)y_1$. Thus, $J_1 \in \mathcal{L}(X, Y)$ and $J_1 z_1 = 1 \cdot y_1 = y_1$.

Set $z_i := T^{L_i} x_i$, for $i \geq 2$ satisfying (16). Suppose that for some $i \geq 2$, the operator $J_{i-1} \in \mathcal{L}(X, Y)$ has already been constructed so that

$$J_{i-1} z_j = y_j, \quad j = 1, \dots, i-1.$$

By (16), we have $z_i \notin \text{span}\{z_1, \dots, z_{i-1}\}$. Let $M_i := \text{span}\{z_1, \dots, z_{i-1}, z_i\}$. Define a linear functional $g_i : M_i \rightarrow \mathbb{R}$ by $g_i(z_j) = 0$, $j = 1, \dots, i-1$, $g_i(z_i) = 1$, and extending linearly on M_i . This is well-defined

because $z_i \notin \text{span}\{z_1, \dots, z_{i-1}\}$. Since M_i is finite-dimensional, g_i is continuous. By the Hahn–Banach Theorem (Conway, 1990, Corollary 3.15, p.112), there exists a bounded linear functional $f_i \in X^*$ extending g_i , hence $f_i(z_j) = 0$, $j < i$, $f_i(z_i) = 1$. Define

$$J_i v := J_{i-1} v + f_i(v)(y_i - J_{i-1} z_i), \quad v \in X.$$

Equivalently, $J_i = J_{i-1} + (y_i - J_{i-1} z_i) \otimes f_i$. Since J_{i-1} and f_i are bounded, $J_i \in \mathcal{L}(X, Y)$.

Now, $J_i z_i = J_{i-1} z_i + f_i(z_i)(y_i - J_{i-1} z_i) = J_{i-1} z_i + (y_i - J_{i-1} z_i) = y_i$. For $j < i$, we have $f_i(z_j) = 0$, so $J_i z_j = J_{i-1} z_j + f_i(z_j)(y_i - J_{i-1} z_i) = J_{i-1} z_j = y_j$. Thus $J_i z_j = y_j$ for every $j \leq i$. By induction, after n_0 steps, for $1 \leq i \leq n_0$: $J_i z_j = y_j$ for $j \leq i$. Finally, by setting $J := J_{n_0} \in \mathcal{L}(X, Y)$ we have $J z_i = J T^{L_i} x_i = y_i$, $i = 1, \dots, n_0$. $\square$

Remark A.3 (Invertible interpolation when $X = Y$). *In the special case $X = Y$, the interpolation map may be chosen as an invertible perturbation of the identity. Indeed, if the recursive construction is initialized with $J_0 = I_X$, then*

$$J = I_X + \sum_{i=1}^N (y_i - J_{i-1} T^{L_i} x_i) \otimes f_i.$$

Hence, if $\sum_{i=1}^N \|y_i - J_{i-1} T^{L_i} x_i\| \|f_i\| < 1$, then $\|J - I_X\| < 1$, so J is an isomorphism by the Neumann lemma (Conway, 1990, Lemma 2.1, p 192). Thus, when $X = Y$, exact interpolation can be realized by an isomorphism $J = I_X + F$, where F has finite rank.

Proof of Theorem 2.1 (Generic orbit independence). We first prove that Ω_D is open and dense in Ω . For each $w = (w_1, \dots, w_d) \in \Omega$, define the $d \times n$ matrix $M(w) := [W_w^{L_1} x_1 \ \dots \ W_w^{L_n} x_n]$, whose i -th column is $W_w^{L_i} x_i$. Note that each coordinate of $W_w^{L_i} x_i$ is a polynomial function of w . Hence every entry of $M(w)$ is polynomial in w .

Now observe that, by definition,

$$w \in \Omega_{\Xi} \iff W_w^{L_i} x_i \notin \text{span}\{W_w^{L_1} x_1, \dots, W_w^{L_{i-1}} x_{i-1}\} \text{ for all } i = 1, \dots, n.$$

This is equivalent to saying that the ordered family $W_w^{L_1} x_1, \dots, W_w^{L_n} x_n$ is linearly independent, and therefore equivalent to the columns of $M(w)$ being linearly independent. Since Ω_{Ξ} is assumed nonempty, there exists at least one w for which these n vectors are linearly independent in $\mathbb{R}^d$; in particular, necessarily $n \leq d$. Therefore, for $w \in \Omega_{\Xi}$, column independence of $M(w)$ is equivalent to the condition $\text{rank } M(w) = n$, which in turn is equivalent to the nonvanishing of at least one $n \times n$ minor of $M(w)$.

For each $w \in \Omega_{\Xi}$, denote $\mathcal{M}_{n,w}$ the collection of all $n \times n$ minors of $M(w)$ (hence polynomials), and set $\mathcal{M}_n := \cup_{w \in \Omega_{\Xi}} \mathcal{M}_{n,w}$. Therefore Ω_{Ξ} can be written as

$$\Omega_{\Xi} = \bigcup_{p \in \mathcal{M}_n} \{w \in \Omega : p(w) \neq 0\},$$

Since each $p \in \mathcal{M}_n$ is a polynomial, each set $\{w \in \Omega : p(w) \neq 0\}$ is open in Ω . Indeed, for any polynomial p , the set $\{w : p(w) \neq 0\}$ is the complement of the zero set $\{w : p(w) = 0\}$. Since polynomials are continuous, zero sets are closed, hence nonvanishing sets are open. Therefore, every set of the form $\{w : p(w) \neq 0\}$ is open in the relative topology of Ω , and any union of such sets is open. So Ω_{Ξ} is open.

To prove density, use again the hypothesis $\Omega_{\Xi} \neq \emptyset$. Thus there exists $w_0 \in \Omega_{\Xi}$ such that the columns of $M(w_0)$ are linearly independent. Equivalently, at least one $n \times n$ minor of $M(w_0)$ is nonzero. Let p denote such a minor. Then $p(w_0) \neq 0$, so p is not the zero polynomial.

Since p is a nonzero polynomial on $\mathbb{R}^d$, its zero set $Z(p) := \{w \in \mathbb{R}^d : p(w) = 0\}$ has empty interior. Equivalently, the set $\{w \in \mathbb{R}^d : p(w) \neq 0\}$ is dense in $\mathbb{R}^d$. Intersecting with $\Omega = (\mathbb{R} \setminus \{0\})^d$, which is open in $\mathbb{R}^d$, we conclude that $\{w \in \Omega : p(w) \neq 0\}$ is dense in Ω . Since

$$\{w \in \Omega : p(w) \neq 0\} \subset \Omega_{\Xi},$$

it follows that Ω_Ξ is dense in Ω .

To prove the measure-zero statement, let p be the nonzero $n \times n$ minor chosen above. Since

$$\{w \in \Omega : p(w) \neq 0\} \subseteq \Omega_D,$$

we have

$$\Omega \setminus \Omega_D \subseteq \{w \in \Omega : p(w) = 0\} \subseteq Z(p).$$

Because p is a nonzero polynomial on $\mathbb{R}^d$, its zero set $Z(p)$ has Lebesgue measure zero in $\mathbb{R}^d$. Therefore

$$\text{Leb}(\Omega \setminus \Omega_D) = 0.$$

□

A.2 Proofs for scaffold selection

Prof of Theorem 2.3 (Existence of best scaffold). First, let us compute an upper bound for the dimension of any scaffold U relative to Ξ . Pick any subdataset $\Xi_{\text{sub}} \subset \Xi$ and assume it is partitioned as $\Xi_{\text{sub}} = D_0 \sqcup \dots \sqcup D_{K-1}$. Then, each sample $x_i \in \Xi_{\text{sub}}$ belongs to exactly one block D_k . Hence, in the dictionary construction, x_i contributes only the d atoms $J_k W^m x_i$, $0 \leq m \leq d-1$, associated with its own interpolant J_k . Therefore the total number of generating atoms is at most $d|\Xi_{\text{sub}}|$, and since the scaffold is a subspace of $\mathbb{R}^C$, we obtain $\dim U \leq \min(C, d|\Xi_{\text{sub}}|) \leq \min(C, dn)$. If $\mathcal{U}_k$ denotes the set of all k -dimensional scaffolds relative to Ξ . Hence,

$$\|A - \Pi_{U_*} A\|_\Xi = \min_{1 \leq k \leq \min(C, dn)} \min_{U \in \mathcal{U}_k} \|A - \Pi_U A\|_\Xi.$$

In order to conclude our proof, it will be enough to prove the existence of a best orbit-constrained scaffold of dimension k , for all $1 \leq k \leq \min(C, dn)$. Since Ξ is finite, there are only finitely many admissible choices of subsets and partitions on which the exact interpolation theorem can be applied. Each such admissible choice produces one scaffold. Therefore, for each fixed k , the family of all k -dimensional scaffolds induced by Ξ is finite, hence compact. On the other hand, the map $U \mapsto \Pi_U$ is continuous (Conway, 1990). Define $f(U) = \|A - \Pi_U A\|_\Xi$. Since A is fixed and $U \mapsto \Pi_U$ is continuous in operator norm, f is continuous on $\mathcal{U}_k$. Because $\mathcal{U}_k$ is compact and f is continuous, the extreme value theorem implies that f attains its minimum on $\mathcal{U}_k$. Hence, there exists $U_*^{(k)} \in \mathcal{U}_k$ such that $f(U_*^{(k)}) = \min_{U \in \mathcal{U}_k} f(U) = \min_{U \in \mathcal{U}_k} \|A - \Pi_U A\|_\Xi$. □

Proof of Proposition 2.4 (One step feasible scaffold expansion.) Since $u \perp U$ and $\|u\|_2 = 1$, $\Pi_{U^+} = \Pi_U + uu^\top$. Hence $(I - \Pi_{U^+})Y = R - u(u^\top Y)$. Since $u \in U^\perp$, $u^\top R = u^\top (I - \Pi_U)Y = u^\top Y$, and therefore $(I - \Pi_{U^+})Y = R - u(u^\top R)$. It follows that

$$\begin{aligned} E^+ &= \|R - u(u^\top R)\|_F^2 \\ &= \|R\|_F^2 - 2\langle R, u(u^\top R) \rangle_F + \|u(u^\top R)\|_F^2 \\ &= E - \|u^\top R\|_2^2. \end{aligned}$$

Thus $E^+ = E - q(u)$.

Since $u \in S$, we have $\Pi_S u = u$, and hence $q(u) = \|u^\top R\|_2^2 = u^\top R R^\top u = u^\top \Pi_S R R^\top \Pi_S u = u^\top M u$. The matrix M is symmetric positive semidefinite. Therefore $q(u) \in [\lambda_{\min}, \lambda_{\max}]$ for every unit vector $u \in S$.

It remains to show that every value in this interval is attained. If $\dim S = 1$, the unit sphere of S consists of two antipodal points and $q(-u) = q(u)$, so $\lambda_{\min} = \lambda_{\max}$ and the interval is a singleton.

Assume $\dim S \geq 2$. If $\lambda_{\max} = \lambda_{\min}$, the conclusion is again immediate. Otherwise, let $v_{\max}$ and $v_{\min}$ be orthonormal unit eigenvectors associated with $\lambda_{\max}$ and $\lambda_{\min}$. Given $e \in [\lambda_{\min}, \lambda_{\max}]$, set

$a := \sqrt{\frac{e - \lambda_{\min}}{\lambda_{\max} - \lambda_{\min}}}$, $b := \sqrt{\frac{\lambda_{\max} - e}{\lambda_{\max} - \lambda_{\min}}}$, and define $u_e := av_{\max} + bv_{\min}$. Since $a^2 + b^2 = 1$ and $v_{\max} \perp v_{\min}$, $\|u_e\|_2 = 1$. Moreover,

$$\begin{aligned} q(u_e) &= u_e^\top M u_e \\ &= a^2 \lambda_{\max} + b^2 \lambda_{\min} \\ &= \frac{e - \lambda_{\min}}{\lambda_{\max} - \lambda_{\min}} \lambda_{\max} + \frac{\lambda_{\max} - e}{\lambda_{\max} - \lambda_{\min}} \lambda_{\min} \\ &= e. \end{aligned}$$

Thus every $e \in [\lambda_{\min}, \lambda_{\max}]$ is achievable, and the corresponding residual energy is $E^+ = E - e$. $\square$

Proof of Theorem 2.5 (Constructive deviation control scaffold). We proceed by induction on t . For $t = 0$, by definition, $U_0 = \{0\}$, $R_0 = Y$, $\|R_0\|_F^2 = E_0$.

Assume that for some $t \in \{1, \dots, T\}$ we have constructed $U_{t-1} \subseteq V$ such that $R_{t-1} := (I - \Pi_{U_{t-1}})Y$ and $\|R_{t-1}\|_F^2 = E_{t-1}$. By Proposition 2.4, the achievable one-step drops obtained by adjoining a unit vector from $S_{t-1} = V \cap U_{t-1}^\perp$ are exactly $[\lambda_{\min}^{(t-1)}, \lambda_{\max}^{(t-1)}]$. Since $E_{t-1} - E_t \in [\lambda_{\min}^{(t-1)}, \lambda_{\max}^{(t-1)}]$, there exists a unit vector $u_t \in S_{t-1}$ such that $q(u_t) = E_{t-1} - E_t$. Define $U_t := U_{t-1} \oplus \text{span}\{u_t\}$. Since $u_t \in V$ and $U_{t-1} \subseteq V$, we have $U_t \subseteq V$. Moreover, $u_t \perp U_{t-1}$, so the chain remains nested.

Setting $R_t := (I - \Pi_{U_t})Y$, Proposition 2.4 gives $\|R_t\|_F^2 = \|R_{t-1}\|_F^2 - q(u_t) = E_{t-1} - (E_{t-1} - E_t) = E_t$. Thus the induction closes, and $\|R_t\|_F^2 = E_t$ for every $t = 1, \dots, T$.

Whenever $\dim S_{t-1} \geq 2$, Proposition 2.4 gives an explicit choice of u_t realizing the prescribed drop $E_{t-1} - E_t$. $\square$

A.3 Proofs for CCDS

The proof of Theorem 2.6 relies on the following elementary Hilbert-space calculation, which characterizes exactly the feasible values of the one-step CCDS deviation.

Lemma A.4 (Distance equation: explicit solutions and feasibility). *Let $Y \subset \mathcal{H}$ be closed, let $u \in Y^\perp$ with $\|u\| = 1$, and let $r \in \mathcal{H}$. Set $a := (I - \Pi_Y)r \in Y^\perp$, $\alpha := \langle a, u \rangle$. Then, for any $d \geq 0$, the equation $\text{dist}(r - \lambda u, Y) = d$ has a real solution $\lambda \in \mathbb{R}$ if and only if $d \geq d_{\min} := \inf_{\lambda \in \mathbb{R}} \text{dist}(r - \lambda u, Y) = \sqrt{\|a\|^2 - \alpha^2}$. When feasible, the solutions are $\lambda_\pm = \alpha \pm \sqrt{d^2 - d_{\min}^2}$. Moreover, the unique minimizer is $\lambda^* = \alpha$, attaining $\text{dist}(r - \lambda^* u, Y) = d_{\min}$.*

Proof. Since $u \in Y^\perp$ and $\|u\| = 1$,

$$\text{dist}(r - \lambda u, Y)^2 = \|(I - \Pi_Y)(r - \lambda u)\|^2 = \|a - \lambda u\|^2 = \|a\|^2 - 2\lambda \langle a, u \rangle + \lambda^2 = (\lambda - \alpha)^2 + (\|a\|^2 - \alpha^2).$$

Hence $\text{dist}(r - \lambda u, Y) = d$ is equivalent to $(\lambda - \alpha)^2 = d^2 - (\|a\|^2 - \alpha^2)$. A real solution therefore exists if and only if $d^2 \geq \|a\|^2 - \alpha^2$, equivalently $d \geq d_{\min}$, and in that case $\lambda = \alpha \pm \sqrt{d^2 - d_{\min}^2}$. Finally, the squared distance is minimized uniquely at $\lambda^* = \alpha$, where its value is $d_{\min}^2$. $\square$

Proof of Theorem 2.6 (Controlled deviation principle). We proceed sequentially in t .

Suppose that $\lambda_1, \dots, \lambda_{t-1}$ have already been chosen, so that $\hat{y}_{t-1} \in Y_{t-1}$ and $r_{t-1} = y - \hat{y}_{t-1}$ are determined. Since $u_t \in Y_t \cap Y_{t-1}^\perp$ and $\|u_t\| = 1$, set

$$a_t := (I - \Pi_{Y_{t-1}})r_{t-1}, \quad \alpha_t := \langle a_t, u_t \rangle.$$

By Lemma A.4, the equation

$$\text{dist}(r_{t-1} - \lambda u_t, Y_{t-1}) = d_t$$

admits a real solution if and only if

$$d_t \geq \sqrt{\|a_t\|^2 - \alpha_t^2} = d_{t,\min}.$$

Under the assumed feasibility condition, its solutions are precisely

$$\lambda = \alpha_t \pm \sqrt{d_t^2 - d_{t,\min}^2}.$$

Choose one of these two roots according to the prescribed deterministic rule and denote it by λ_t . Define

$$\hat{y}_t := \hat{y}_{t-1} + \lambda_t u_t, \quad r_t := y - \hat{y}_t.$$

Since $\hat{y}_{t-1} \in Y_{t-1} \subset Y_t$ and $u_t \in Y_t$, one has $\hat{y}_t \in Y_t$. Moreover,

$$r_t = y - \hat{y}_{t-1} - \lambda_t u_t = r_{t-1} - \lambda_t u_t.$$

Therefore, by the defining choice of λ_t ,

$$\text{dist}(r_t, Y_{t-1}) = \text{dist}(r_{t-1} - \lambda_t u_t, Y_{t-1}) = d_t.$$

Starting from $\hat{y}_0 = 0$ and $r_0 = y$, this construction can be applied successively for $t = 1, \dots, T$. Hence every feasible profile $(d_t)_{t=1}^T$ is realized, and the resulting coefficients satisfy

$$\lambda_t = \alpha_t \pm \sqrt{d_t^2 - d_{t,\min}^2}, \quad t = 1, \dots, T.$$

The deterministic root-selection rule makes the resulting sequence $(\lambda_t)_{t=1}^T$ deterministic. $\square$

Explicit computation of the CCDS dictionary-coordinate vector. Let $q_1, \dots, q_T$ be the atoms selected along the CCDS path, with u_t and Y_t defined as in (8). Let $\eta_t := \|q_t - \Pi_{Y_{t-1}} q_t\| > 0$, and let $e_t \in \mathbb{R}^t$ denote the t -th canonical basis vector. Define recursively coefficient vectors $c^{(t)} \in \mathbb{R}^t$ by $c^{(1)} := (1)$, and, for $t \geq 2$,

$$c^{(t)} := \frac{1}{\eta_t} \left(e_t - \sum_{s=1}^{t-1} \langle q_t, u_s \rangle \tilde{c}^{(s,t)} \right), \quad (17)$$

where $\tilde{c}^{(s,t)} \in \mathbb{R}^t$ is the zero-padded extension of $c^{(s)}$, namely $\tilde{c}^{(s,t)} := (c_1^{(s)}, \dots, c_s^{(s)}, 0, \dots, 0)$. Set $\beta^{(0)} := 0 \in \mathbb{R}^0$, and recursively define

$$\beta^{(t)} := (\beta^{(t-1)}, 0) + \lambda_t c^{(t)} \in \mathbb{R}^t. \quad (18)$$

Proposition A.5 (Explicit CCDS dictionary-coordinate recursion). *For each $t = 1, \dots, T$ one has $u_t = \sum_{j=1}^t c_j^{(t)} q_j$, and consequently $\hat{y}_T = \sum_{t=1}^T \lambda_t u_t = \sum_{j=1}^T \beta_j^{(T)} q_j$.*

Proof. We argue by induction on t . For $t = 1$, since $Y_0 = \{0\}$, one has $u_1 = q_1$, so $c^{(1)} = (1)$ and the claim holds. Assume now that for each $s < t$, $u_s = \sum_{j=1}^s c_j^{(s)} q_j$. Since $\{u_s\}_{s=1}^{t-1}$ is an orthonormal basis of Y_{t-1} , we have $\Pi_{Y_{t-1}} q_t = \sum_{s=1}^{t-1} \langle q_t, u_s \rangle u_s$. Substituting the inductive representation of each u_s ,

$$q_t - \Pi_{Y_{t-1}} q_t = q_t - \sum_{s=1}^{t-1} \langle q_t, u_s \rangle \sum_{j=1}^s c_j^{(s)} q_j.$$

After zero-padding the vectors $c^{(s)}$ to $\mathbb{R}^t$, this becomes

$$q_t - \Pi_{Y_{t-1}} q_t = \sum_{j=1}^t \left(e_t - \sum_{s=1}^{t-1} \langle q_t, u_s \rangle \tilde{c}^{(s,t)} \right)_j q_j.$$

Dividing by $\eta_t = \|q_t - \Pi_{Y_{t-1}} q_t\|$ yields $u_t = \sum_{j=1}^t c_j^{(t)} q_j$. This proves the first claim. Finally,

$$\hat{y}_T = \sum_{t=1}^T \lambda_t u_t = \sum_{t=1}^T \lambda_t \sum_{j=1}^t c_j^{(t)} q_j = \sum_{j=1}^T \beta_j^{(T)} q_j,$$

by the recursive definition of $\beta^{(t)}$. This proves the result. $\square$

Proof of Proposition 2.7 (Minimal deviation CCDS recovers orthogonal coordinates). Starting from $Y_0 = \{0\}$, define for $t = 1, \dots, s$, as in (8),

$$\eta_t := \|q_t - \Pi_{Y_{t-1}} q_t\| > 0, \quad u_t := \frac{q_t - \Pi_{Y_{t-1}} q_t}{\eta_t}, \quad Y_t := Y_{t-1} \oplus \text{span}\{u_t\}.$$

Since $q_1, \dots, q_s$ are linearly independent, $q_t \notin Y_{t-1}$ for every t , hence $\eta_t > 0$ and each u_t is well defined. By construction, $u_t \perp Y_{t-1}$, $\|u_t\| = 1$, and therefore $Y_t = \text{span}\{q_1, \dots, q_t\} = \text{span}\{u_1, \dots, u_t\}$. Thus $u_1, \dots, u_s$ is an orthonormal basis of E . Write the orthonormal expansion of $\Pi_E y$ as $\Pi_E y = \sum_{j=1}^s \gamma_j u_j$, with $\gamma_j := \langle \Pi_E y, u_j \rangle$. Set $\text{CCDS}_{\mathcal{D}}^t(y; q_{1:t}, d_{\min, 1:t}) := \hat{y}_t$. We prove by induction that

$$\hat{y}_{t-1} = \sum_{j=1}^{t-1} \gamma_j u_j, \quad r_{t-1} = \sum_{j=t}^s \gamma_j u_j.$$

For $t = 1$, this follows from $\hat{y}_0 = 0$ and $r_0 = \Pi_E y$. Assume the claim holds before step t . Since $Y_{t-1} = \text{span}\{u_1, \dots, u_{t-1}\}$ and r_{t-1} is a linear combination of $u_t, \dots, u_s$, we have $r_{t-1} \perp Y_{t-1}$. Hence the CCDS off-scaffold residual is

$$a_t := (I - \Pi_{Y_{t-1}}) r_{t-1} = r_{t-1}.$$

Therefore

$$\alpha_t := \langle a_t, u_t \rangle = \left\langle \sum_{j=t}^s \gamma_j u_j, u_t \right\rangle = \gamma_t.$$

Choosing $d_t = d_{t,\min}$ in the CCDS root formula gives $\lambda_t = \alpha_t \pm \sqrt{d_t^2 - d_{t,\min}^2} = \alpha_t = \gamma_t$. Consequently, $\hat{y}_t = \hat{y}_{t-1} + \lambda_t u_t = \sum_{j=1}^t \gamma_j u_j$, and $r_t = \Pi_E y - \hat{y}_t = \sum_{j=t+1}^s \gamma_j u_j$. This closes the induction. Thus, $\hat{y}_s = \Pi_E y$ and $r_s = 0$. We conclude by Proposition A.5. $\square$

Proof of Proposition 2.8 (MP and OMP as CCDS boundaries). We show that the classical greedy pursuits MP and OMP arise as extremal regimes of the CCDS distance equation, while strict interior CCDS updates are globally inaccessible to OMP, marking a geometric separation between OMP and CCDS.

(i) (MP boundary). For Matching Pursuit (MP), keep the reference space trivial, $Y_{t-1} = \{0\}$, at every step and select $q_t \in \arg \max_{q \in \mathcal{D}} |\langle r_{t-1}, q \rangle|$, that is, use the MP atom-selection rule. Then $u_t = q_t$, $a_t = (I - \Pi_{Y_{t-1}}) r_{t-1} = r_{t-1}$, and $\alpha_t = \langle a_t, u_t \rangle = \langle r_{t-1}, q_t \rangle$. At the minimal-deviation boundary $d_t = d_{t,\min}$, the CCDS root satisfies $\lambda_t = \alpha_t = \langle r_{t-1}, q_t \rangle$, so the CCDS updates become

$$\hat{y}_t = \hat{y}_{t-1} + \lambda_t q_t, \quad r_t = r_{t-1} - \lambda_t q_t,$$

which are exactly the Matching Pursuit iterates.

(ii) (OMP boundary). In contrast, when the previously selected directions are retained through the nested spaces $Y_t = \text{span}\{q_1, \dots, q_t\}$, Proposition 2.7 shows that, for any fixed linearly independent atom ordering $q_{1:T}$, the choice $d_t = d_{t,\min}$ gives the CCDS output $\hat{y}_t = \Pi_{Y_t} y$. Hence, if q_t is selected at each step by the OMP atom-selection rule, minimal-deviation CCDS and OMP produce the same iterates until termination. Thus MP and OMP correspond respectively to the minimal-deviation boundary without and with accumulated subspace geometry, whereas full CCDS retains the nested geometry while allowing the interior regime $d_t > d_{t,\min}$.

(iii) (Global geometric separation from OMP). First, we establish a global necessary condition for every OMP iterate, independent of the atom ordering. Note that every OMP iterate is the orthogonal projection of y onto some dictionary-generated selected subspace Z : $\hat{y}^{\text{OMP}} = \Pi_Z y$. Hence $y - \hat{y}^{\text{OMP}} \perp Z$, and $\hat{y}^{\text{OMP}} \in Z$, and therefore every OMP iterate satisfies the global invariant

$$\langle y - \hat{y}^{\text{OMP}}, \hat{y}^{\text{OMP}} \rangle = 0. \quad (19)$$

Now, in order to establish the global separation result, we show that any interior CCDS iterate violates the invariant (19). Assume we depart from an OMP boundary state $\hat{y}_{t-1} = \Pi_{Y_{t-1}} y$, so we have $r_{t-1} =$

$y - \Pi_{Y_{t-1}} y \in Y_{t-1}^\perp$, and therefore

$$a_t = (I - \Pi_{Y_{t-1}})r_{t-1} = r_{t-1}.$$

Let $u_t \in Y_{t-1}^\perp$ be unit norm, then

$$\alpha_t = \langle a_t, u_t \rangle = \langle r_{t-1}, u_t \rangle = \langle y - \Pi_{Y_{t-1}} y, u_t \rangle = \langle y, u_t \rangle.$$

Consider the CCDS update $\hat{y}_t^{\text{CCDS}} = \hat{y}_{t-1} + \lambda_t u_t$. Since $y - \hat{y}_{t-1} \perp Y_{t-1}$ and $u_t \perp Y_{t-1}$,

$$\langle y - \hat{y}_t^{\text{CCDS}}, \hat{y}_t^{\text{CCDS}} \rangle = \langle (y - \hat{y}_{t-1}) - \lambda_t u_t, \hat{y}_{t-1} + \lambda_t u_t \rangle = \lambda_t \langle y, u_t \rangle - \lambda_t^2 = \lambda_t(\alpha_t - \lambda_t).$$

Thus, whenever $\lambda_t \notin \{0, \alpha_t\}$, we have $\langle y - \hat{y}_t^{\text{CCDS}}, \hat{y}_t^{\text{CCDS}} \rangle \neq 0$. Hence $\hat{y}_t^{\text{CCDS}}$ violates the invariant (19) satisfied by every OMP iterate and therefore cannot be produced by OMP from any selected dictionary subspace.

For a strict CCDS interior choice $d_t > d_{t,\min}$, one has

$$\lambda_t = \alpha_t \pm \sqrt{d_t^2 - d_{t,\min}^2} \neq \alpha_t;$$

apart from the exceptional zero-update case $\lambda_t = 0$, the resulting CCDS iterate is therefore globally inaccessible to OMP. $\square$

Finite-dimensional CCDS feasibility obstruction and lifting. At step t , the prescribed deviation is feasible in the current ambient space if and only if $d_t \geq d_{t,\min}$. Hence any choice $d_t < d_{t,\min}$ is geometrically unrealizable: no scalar λ_t can satisfy $\text{dist}(r_{t-1} - \lambda_t u_t, Y_{t-1}) = d_t$. In finite dimension this obstruction can arise when the available innovation direction is nearly orthogonal to the off-scaffold residual, motivating dimension lifting to restore feasibility.

Example A.6 (Finite-dimensional infeasibility). *Let $H = \mathbb{R}^2$, $Y_0 = \{0\}$, and take the initial residual $r_0 = e_1 = (1, 0)$. Suppose the only available innovation direction is $u_1 = e_2 = (0, 1)$. Since $Y_0 = \{0\}$, we have $a_1 = (I - \Pi_{Y_0})r_0 = e_1$ and hence $\alpha_1 = \langle a_1, u_1 \rangle = 0$. Therefore $d_{1,\min} = \sqrt{\|a_1\|^2 - \alpha_1^2} = 1$. Thus no prescribed deviation $d_1 < 1$ is feasible: there is no $\lambda \in \mathbb{R}$ such that $\text{dist}(r_0 - \lambda u_1, Y_0) = d_1$. The obstruction comes from the available innovation direction being orthogonal to the residual; dimension lifting restores feasibility by introducing a direction with nonzero alignment with it.*

Adding one new orthogonal dimension restores feasibility by allowing a new innovation direction with controlled alignment to a . This justifies the need for dimension lifting in order to guarantee universal deviation feasibility and, consequently, collapse of the parametrization error in CCDS.

Theorem A.7 (One-step feasibility by dimension lifting). *Let H be a real Hilbert space, $Y \subset H$ a closed subspace, and $r \in H$. Define $a := (I - \Pi_Y)r \in Y^\perp$. Fix a target deviation d satisfying $0 \leq d \leq \|a\|$. Then there exists a Hilbert space $\tilde{H} := H \oplus \mathbb{R}$, an isometric embedding $\iota : H \rightarrow \tilde{H}$, and a unit vector $u \in (\iota(Y))^\perp \subset \tilde{H}$ such that the CCDS distance equation $\text{dist}(\iota(r) - \lambda u, \iota(Y)) = d$ admits a real solution $\lambda \in \mathbb{R}$. Moreover, one can choose u so that the minimal achievable deviation is exactly $d_{\min}(u) = d$.*

Proof. Define $\tilde{H} := H \oplus \mathbb{R}$ with inner product

$$\langle (x, s), (y, t) \rangle_{\tilde{H}} := \langle x, y \rangle_H + st,$$

and the isometric embedding $\iota(x) = (x, 0)$. Then $\iota(Y)$ is a closed subspace and

$$(\iota(Y))^\perp = Y^\perp \oplus \mathbb{R}.$$

Let

$$\tilde{r} := \iota(r), \quad \tilde{a} := (I - \Pi_{\iota(Y)})\tilde{r} = (a, 0),$$

so that $\|\tilde{a}\| = A$. If $A = 0$, then $d = 0$ and the claim is trivial. Assume $A > 0$ and define $e_1 := \frac{\tilde{a}}{A}$, $e_2 := (0, 1)$. Then $e_1, e_2 \in (\iota(Y))^\perp$ are orthonormal. Choose $\theta \in [0, \pi/2]$ such that $\sin \theta = \frac{d}{A}$. Define $u := \cos \theta e_1 + \sin \theta e_2$. Then $u \in (\iota(Y))^\perp$ and $\|u\| = 1$. Moreover, $\alpha := \langle \tilde{a}, u \rangle_{\tilde{H}} = A \cos \theta$.

By the CCDS feasibility formula,

$$d_{\min}(u) = \sqrt{\|\tilde{a}\|^2 - \alpha^2} = \sqrt{A^2 - A^2 \cos^2 \theta} = A \sin \theta = d.$$

Hence $d \geq d_{\min}(u)$, and the equation $\text{dist}(\tilde{r} - \lambda u, \iota(Y)) = d$ admits a real solution λ . This completes the proof. $\square$

Remark A.8. To enforce the same target deviation level d simultaneously for M sample-wise residuals $\{r^{(j)}\}_{j=1}^M$ at a given CCDS step, it suffices to iterate the one-step dimension-lifting construction of Theorem A.7 for each residual. This yields an isometric embedding into an ambient Hilbert space of dimension at most $\dim(H) + M$ in which the deviation constraint $\text{dist}(r^{(j)} - \lambda u, Y) = d$ is feasible for all $j = 1, \dots, M$.

Resolution of Example A.6 by lifting one dimension. Embed H isometrically into $\tilde{H} = \mathbb{R}^3$ by $x \mapsto (x, 0)$, and extend the orthogonal complement with the new basis vector e_3 . Choose the lifted innovation direction $\tilde{u}_t = \cos \theta e_1 + \sin \theta e_3$, $\sin \theta = d_t$. Then $\tilde{u}_t \in (\iota(Y_{t-1}))^\perp$, $\|\tilde{u}_t\| = 1$, and $\langle \tilde{a}, \tilde{u}_t \rangle = \cos \theta$, $d_{\min}(\tilde{u}_t) = \sqrt{1 - \cos^2 \theta} = \sin \theta = d_t$. Hence the CCDS equation

$$\text{dist}(\iota(r_{t-1}(x_1)) - \lambda \tilde{u}_t, \iota(Y_{t-1})) = d_t$$

admits a real solution, and the deviation d_t becomes feasible after adding a single orthogonal dimension.

A.4 Auxiliary local-chart projection estimate

Proof of Theorem 2.11 (Local interpolant as an observable local chart). Let $X := [x_1 \ \cdots \ x_s]$. The compatibility condition (9) implies that $\ker X \subset \ker Y$. Since $X^\dagger X$ is the orthogonal projection of $\mathbb{R}^s$ onto $(\ker X)^\perp$, and since Y vanishes on $\ker X$, we have $Y = YX^\dagger X$. indeed, every $c \in \mathbb{R}^s$ decomposes uniquely as $c = c_0 + c_1$ with $c_0 \in \ker X$ and $c_1 \in (\ker X)^\perp$, then $YX^\dagger Xc = Yc_1 = Y(c_0 + c_1) = Yc$. Therefore the operator $A_D := YX^\dagger : \mathbb{R}^d \rightarrow \mathbb{R}^C$ satisfies $YX^\dagger x_i = YX^\dagger X e_i = Y e_i = y_i$. Moreover,

$$\|A_D\| = \|YX^\dagger\| \leq \|Y\| \|X^\dagger\| = \frac{\|Y\|}{\sigma_{\min}^+(X)} \leq \frac{\|Y\|_F}{\sigma_{\min}^+(X)}. \quad (20)$$

Now define the observable defect matrix

$$\Delta_D^{\text{obs}} := Y - A_D Z = [y_1 - A_D z_1 \ \cdots \ y_s - A_D z_s].$$

Since Z has full column rank, $ZZ^\dagger = \Pi_E$. Therefore

$$J_D - A_D \Pi_E = YZ^\dagger - A_D ZZ^\dagger = (Y - A_D Z)Z^\dagger = \Delta_D^{\text{obs}} Z^\dagger.$$

Taking operator norms gives

$$\|J_D - A_D \Pi_E\| \leq \|\Delta_D^{\text{obs}}\| \|Z^\dagger\| = \frac{\|\Delta_D^{\text{obs}}\|}{\sigma_{\min}(Z)} \leq \frac{\|\Delta_D^{\text{obs}}\|_F}{\sigma_{\min}(Z)}. \quad (21)$$

Finally, $y_i - A_D z_i = A_D x_i - A_D z_i = A_D(x_i - z_i)$. Thus $\|\Delta_D^{\text{obs}}\|_F^2 = \sum_{i=1}^s \|A_D(x_i - z_i)\|^2 \leq \|A_D\|^2 \sum_{i=1}^s \|x_i - z_i\|^2$. Using this estimate together with (21) and (20) yields (10). $\square$

Theorem A.9 (Local-chart diagnostic before scaffold selection). *Let $\Xi_{\text{sub}} = \{(x_i, y_i)\}_{i=1}^N \subset \mathbb{R}^d \times \mathbb{R}^C$, and let $P = \{D_1, \dots, D_K\}$ be a partition of Ξ_{sub} . Write $D_k = \{(x_{k,1}, y_{k,1}), \dots, (x_{k,s_k}, y_{k,s_k})\}$ and set $X_k = [x_{k,1} \ \cdots \ x_{k,s_k}]$ and $Y_k = [y_{k,1} \ \cdots \ y_{k,s_k}]$. Assume that, for every k , $\sum_{j=1}^{s_k} c_j x_{k,j} = 0$ implies $\sum_{j=1}^{s_k} c_j y_{k,j} = 0$. Choose depths $L_{k,j}$ such that $z_{k,j} := W^{L_{k,j}} x_{k,j}$ are linearly independent, and set $Z_k = [z_{k,1} \ \cdots \ z_{k,s_k}]$, $E_k = \text{span}\{z_{k,1}, \dots, z_{k,s_k}\}$, and $J_k := Y_k Z_k^\dagger$. Let U_P be the orbit scaffold generated by $\{J_k W^m x_{k,j} : 1 \leq k \leq K, 1 \leq j \leq s_k, 0 \leq m \leq d-1\}$. Then for all $1 \leq k \leq K$ and $1 \leq j \leq s_k$, we have the anchor-level scaffold diagnostic*

$$\|(I - \Pi_{U_P} y_{k,j})\| \leq \frac{\|Y_k\|_F}{\sigma_{\min}^+(X_k)} \text{dist}(x_{k,j}, E_k) + \delta_k \|x_{k,j}\|,$$

where $\delta_k := \|J_k - Y_k X_k^\dagger \Pi_{E_k}\|$.

Proof. Fix a block D_k and set $A_k := Y_k X_k^\dagger$. The compatibility assumption gives $\ker X_k \subseteq \ker Y_k$, as in the proof of Theorem 2.11 we have $Y_k = Y_k X_k^\dagger X_k$. Therefore $A_k x_{k,j} = y_{k,j}$ for every j . Also,

$$\|A_k\| = \|Y_k X_k^\dagger\| \leq \|Y_k\|_F \|X_k^\dagger\| = \frac{\|Y_k\|_F}{\sigma_{\min}^+(X_k)}.$$

Since $m = 0$ is allowed in the orbit scaffold, $J_k x_{k,j} \in U_P$. Hence $\text{dist}(y_{k,j}, U_P) \leq \|y_{k,j} - J_k x_{k,j}\|$. Using $y_{k,j} = A_k x_{k,j}$ and inserting $A_k \Pi_{E_k} x_{k,j}$, we get

$$\|y_{k,j} - J_k x_{k,j}\| \leq \|A_k(I - \Pi_{E_k})x_{k,j}\| + \|(J_k - A_k \Pi_{E_k})x_{k,j}\|.$$

Thus

$$\|(I - \Pi_{U_P} y_{k,j})\| = \text{dist}(y_{k,j}, U_P) \leq \frac{\|Y_k\|_F}{\sigma_{\min}^+(X_k)} \text{dist}(x_{k,j}, E_k) + \delta_k \|x_{k,j}\|.$$

□

Proof of Theorem 2.10 (Stability of the scaffold/CCDS bottlenecks under relaxed interpolation). For each $1 \leq k \leq K$, let $D_k = \{(x_{k,1}, y_{k,1}), \dots, (x_{k,|D_k|}, y_{k,|D_k|})\} \subseteq \Xi_{\text{sub}}$ and choose depths $L_{k,1}, \dots, L_{k,|D_k|}$ such that $y_{k,j} = J_k W^{(L_{k,j})} x_{k,j}$ for all $(x_{k,j}, y_{k,j}) \in D_k$. Let $\mathcal{I} = \{(k, m, j) : 1 \leq k \leq K, 0 \leq m \leq d-1, 1 \leq j \leq |D_k|\}$ be the fixed finite atom index set of the corresponding scaffold as defined in (4). Write the reference atoms as $g_i = J_k W^{(m)} x_{k,j}$, with $i = (k, m, j) \in \mathcal{I}$, and the perturbed atoms as $g'_i = J'_k W^{(m)} x_{k,j}$. Let $G = [g_i]_{i \in \mathcal{I}}$, $G' = [g'_i]_{i \in \mathcal{I}}$, $U = \text{range}(G)$, $U' = \text{range}(G')$, and $\eta := \|G' - G\|_2$. Also set $V := \sum_{k=1}^K \text{range}(Y_k)$.

Since $J'_k = Y_k R'_k$, every perturbed atom g'_i belongs to $\text{range}(Y_k)$, hence $U' \subseteq V$. On the other hand, suppose $y \in V$, then there exist scalars $(\beta_{k,j})_{1 \leq k \leq K, 1 \leq j \leq |D_k|}$ such that $y = \sum_{k=1}^K \sum_{j=1}^{|D_k|} \beta_{k,j} y_{k,j} = \sum_{k=1}^K \sum_{j=1}^{|D_k|} \beta_{k,j} J_k W^{(L_{k,j})} x_{k,j} \in U$. Therefore $V \subseteq U$. Conversely, every reference atom is of the form $g_i = Y_k Z_k^\dagger W^{(m)} x_{k,j}$, hence belongs to V . Thus $U = V$, and consequently $U' \subseteq U$.

Let $r = \text{rank}(G) = \dim U$ and $\gamma = \sigma_r(G) > 0$. Choose $\delta_0 < \gamma/2$. If $\eta < \delta_0$, then Weyl's singular-value perturbation inequality (see Horn & Johnson, 2012) gives $\sigma_r(G') \geq \sigma_r(G) - \eta > \gamma/2 > 0$. Hence $\text{rank}(G') \geq r$. Since $U' \subseteq U$ and $\dim U = r$, also $\text{rank}(G') \leq r$. Therefore $\text{rank}(G') = r$, and so $U' = U$. It follows immediately that $\Pi_{U'} = \Pi_U$, and hence $\varepsilon_{\text{proj}}(A, U'; \Xi) = \varepsilon_{\text{proj}}(A, U; \Xi)$.

It remains to prove the parametrization bound. Since $U' = U$, the projected targets are identical: $b_l := \Pi_U A x_l = \Pi_{U'} A x_l$. Fix l and suppress the index l . Let $r_0 = b_l$, and let r_t , $t = 0, \dots, T$, be the residuals of the reference CCDS path of the CCDS realizer f_{CCDS} . At step t , the reference path uses a prescribed atom label α_t , prescribed deviation d_t , and prescribed root branch $\varepsilon_t \in \{-1, 1\}$. Let $Y_{t-1} := \text{span}\{u_1, \dots, u_{t-1}\}$, $P_{t-1} := \Pi_{Y_{t-1}}$, $q_t := (I - P_{t-1})g_t$, $u_t := q_t / \|q_t\|$, $a_t := (I - P_{t-1})r_{t-1}$, $\alpha_t := \langle a_t, u_t \rangle$, and $d_{t,\min}^2 := \|a_t\|^2 - \alpha_t^2$.

The CCDS coefficient is $\lambda_t = \alpha_t + \varepsilon_t \sqrt{d_t^2 - d_{t,\min}^2}$, and the residual update is $r_t = r_{t-1} - \lambda_t u_t$.

We transport exactly this finite prescription to the perturbed atoms: the same labels i_t , the same deviations d_t , and the same root branches ε_t . Thus g_{i_t} is replaced by the corresponding g'_{i_t} , and all perturbed quantities Y'_{t-1} , P'_{t-1} , q'_t , u'_t , a'_t , α'_t , λ'_t , and r'_t are defined by the same formulas.

We prove by induction that $\|r'_t - r_t\| \leq B_t \eta$ and $\|u'_t - u_t\| \leq A_t \eta$ for constants depending only on the reference trajectory. The base residual estimate is $\|r'_0 - r_0\| = 0$, because $b'_l = b_l$. Assume the estimates are known up to step $t-1$. Since the directions $u_1, \dots, u_{t-1}$ and $u'_1, \dots, u'_{t-1}$ are orthonormal along the CCDS innovation construction, if $U_{t-1} := [u_1 \dots u_{t-1}]$ and $U'_{t-1} := [u'_1 \dots u'_{t-1}]$, then $P_{t-1} = U_{t-1} U_{t-1}^\top$ and $P'_{t-1} = U'_{t-1} (U'_{t-1})^\top$. Add and subtract $U_{t-1} (U'_{t-1})^\top$: $P'_{t-1} - P_{t-1} = (U'_{t-1} - U_{t-1})(U'_{t-1})^\top + U_{t-1} ((U'_{t-1})^\top - U_{t-1}^\top)$. Taking operator norms, $\|P'_{t-1} - P_{t-1}\|_{\text{op}} \leq \|U'_{t-1} - U_{t-1}\|_2 \|(U'_{t-1})^\top\|_2 + \|U_{t-1}\|_2 \|(U'_{t-1})^\top - U_{t-1}^\top\|_2$. Since U_{t-1} and U'_{t-1} have orthonormal columns, $\|U_{t-1}\|_2 = \|U'_{t-1}\|_2 = 1$. Thus $\|P'_{t-1} - P_{t-1}\|_{\text{op}} \leq 2\|U'_{t-1} - U_{t-1}\|_2 \leq 2\|U'_{t-1} - U_{t-1}\|_F = 2\left(\sum_{s=1}^{t-1} \|u'_s - u_s\|^2\right)^{1/2}$. Now the induction hypothesis says that for all previous steps

$s < t$, $\|u'_s - u_s\| \leq A_s \eta$ for some constants A_s depending only on the reference trajectory. Since there are only finitely many steps, define $A_{t-1}^{\max} := \max_{1 \leq s < t} A_s$. Then $\left(\sum_{s=1}^{t-1} \|u'_s - u_s\|^2\right)^{1/2} \leq \sqrt{t-1} A_{t-1}^{\max} \eta$. Hence

$$\|P'_{t-1} - P_{t-1}\|_{\text{op}} \leq 2\sqrt{t-1} A_{t-1}^{\max} \eta = C_{P,t} \eta. \quad (22)$$

Since corresponding atom columns differ by at most η , we have $\|g'_{i_t} - g_{i_t}\| \leq \eta$. Therefore $q'_t - q_t = (I - P'_{t-1})(g'_{i_t} - g_{i_t}) + (P_{t-1} - P'_{t-1})g_{i_t}$, and so

$$\|q'_t - q_t\| \leq \eta + C_{P,t} \eta \|g_{i_t}\| \leq C_{q,t} \eta. \quad (23)$$

By the nondegeneracy assumption, $\|q_t\| \geq \nu > 0$. After decreasing δ_0 , we may assume $\|q'_t - q_t\| \leq \nu/2$. Then by the reverse triangle inequality, $\|q'_t\| \geq \|q_t\| - \|q'_t - q_t\| \geq \nu - \nu/2 = \nu/2$. So both q and q'_t stay away from zero. Now prove the Lipschitz bound. We write $\frac{q'}{\|q'\|} - \frac{q}{\|q\|} = \frac{q'_t - q_t}{\|q'_t\|} + q_t \left(\frac{1}{\|q'_t\|} - \frac{1}{\|q_t\|} \right)$. Taking norms, $\left\| \frac{q'_t}{\|q'_t\|} - \frac{q_t}{\|q_t\|} \right\| \leq \frac{\|q'_t - q_t\|}{\|q'_t\|} + \|q_t\| \left| \frac{1}{\|q'_t\|} - \frac{1}{\|q_t\|} \right|$. For the second term, $\left| \frac{1}{\|q'_t\|} - \frac{1}{\|q_t\|} \right| = \frac{|\|q_t\| - \|q'_t\||}{\|q'_t\| \|q_t\|}$. By the reverse triangle inequality, $|\|q_t\| - \|q'_t\|| \leq \|q_t - q'_t\|$. Therefore $\|q_t\| \left| \frac{1}{\|q'_t\|} - \frac{1}{\|q_t\|} \right| \leq \|q_t\| \frac{\|q_t - q'_t\|}{\|q'_t\| \|q_t\|} = \frac{\|q_t - q'_t\|}{\|q'_t\|}$. So $\left\| \frac{q'_t}{\|q'_t\|} - \frac{q_t}{\|q_t\|} \right\| \leq \frac{2\|q'_t - q_t\|}{\|q'_t\|}$. Since $\|q'_t\| \geq \nu/2$, we get $\frac{1}{\|q'_t\|} \leq \frac{2}{\nu}$. Hence $\|u'_t - u_t\| = \left\| \frac{q'_t}{\|q'_t\|} - \frac{q_t}{\|q_t\|} \right\| \leq \frac{4}{\nu} \|q'_t - q_t\|$. And by (23) we have

$$\|u'_t - u_t\| \leq A_t \eta. \quad (24)$$

Now compare the CCDS coefficients. We have $a'_t - a_t = (I - P'_{t-1})(r'_{t-1} - r_{t-1}) + (P_{t-1} - P'_{t-1})r_{t-1}$. Since the reference residuals are bounded by (22) it follows that

$$\|a'_t - a_t\| \leq C_{a,t} (\|r'_{t-1} - r_{t-1}\| + \eta). \quad (25)$$

Also $\alpha'_t - \alpha_t = \langle a'_t - a_t, u'_t \rangle + \langle a_t, u'_t - u_t \rangle$, hence

$$|\alpha'_t - \alpha_t| \leq C_{\alpha,t} (\|r'_{t-1} - r_{t-1}\| + \eta). \quad (26)$$

Recall $d_{t,\min}^2 = \|a_t\|^2 - \alpha_t^2$. Similarly, $(d'_{t,\min})^2 = \|a'_t\|^2 - (\alpha'_t)^2$. Then, $(d'_{t,\min})^2 - d_{t,\min}^2 = (\|a'_t\|^2 - \|a_t\|^2) - ((\alpha'_t)^2 - \alpha_t^2)$. Taking absolute values, $|(d'_{t,\min})^2 - d_{t,\min}^2| \leq |\|a'_t\|^2 - \|a_t\|^2| + |(\alpha'_t)^2 - \alpha_t^2|$. Now $|\|a'_t\|^2 - \|a_t\|^2| = |(\|a'_t\| - \|a_t\|)(\|a'_t\| + \|a_t\|)|$. Using $|\|a'_t\| - \|a_t\|| \leq \|a'_t - a_t\|$, we get $|\|a'_t\|^2 - \|a_t\|^2| \leq (\|a'_t\| + \|a_t\|)\|a'_t - a_t\|$. Similarly, $|(\alpha'_t)^2 - \alpha_t^2| = |\alpha'_t - \alpha_t| |\alpha'_t + \alpha_t|$. Along the finite trajectory, $\|a_t\|$, $\|a'_t\|$, $|\alpha_t|$, and $|\alpha'_t|$ stay bounded for sufficiently small perturbations. Therefore by (25) and (26) we have

$$|(d'_{t,\min})^2 - d_{t,\min}^2| \leq C_{m,t} (\|r'_{t-1} - r_{t-1}\| + \eta). \quad (27)$$

Choose the same deterministic root branch in the reference and perturbed problem, for example the minus root: $\lambda_t = \alpha_t - \sqrt{d_t^2 - d_{t,\min}^2}$, $\lambda'_t = \alpha'_t - \sqrt{(d'_t)^2 - (d'_{t,\min})^2}$. Then $|\lambda'_t - \lambda_t| \leq |\alpha'_t - \alpha_t| + \left| \sqrt{d_t^2 - d_{t,\min}^2} - \sqrt{(d'_t)^2 - (d'_{t,\min})^2} \right|$. By the feasibility-margin assumption, $d_t^2 - d_{t,\min}^2 \geq \eta_0^2 > 0$. So if the perturbation is small enough that $C_{m,t}(\|r'_{t-1} - r_{t-1}\| + \eta) \leq \eta_0^2/2$, then by (27) we have $d_t^2 - (d'_{t,\min})^2 \geq d_t^2 - d_{t,\min}^2 - |(d_t^2 - (d'_{t,\min})^2) - (d_t^2 - d_{t,\min}^2)| \geq \eta_0^2 - \eta_0^2/2 = \eta_0^2/2$. Thus the perturbed distance equation remains strictly feasible. The map $s \mapsto \sqrt{s}$ is Lipschitz on $[\eta_0^2/2, \infty)$. Therefore the square-root Lipschitz estimate gives $|\sqrt{(d_t)^2 - (d'_{t,\min})^2} - \sqrt{d_t^2 - d_{t,\min}^2}| \leq C_{\sqrt{\cdot}} |(d_t)^2 - (d'_{t,\min})^2 - (d_t^2 - d_{t,\min}^2)| = C_{\sqrt{\cdot}} |(d'_{t,\min})^2 - d_{t,\min}^2|$. By (26) and (27) we obtain

$$|\lambda'_t - \lambda_t| \leq C_{\lambda,t} (\|r'_{t-1} - r_{t-1}\| + \eta). \quad (28)$$

Finally, $r'_t - r_t = (r'_{t-1} - r_{t-1}) - (\lambda'_t u'_t - \lambda_t u_t)$, and $\lambda'_t u'_t - \lambda_t u_t = (\lambda'_t - \lambda_t) u'_t + \lambda_t (u'_t - u_t)$. Since $\|u'_t\| = 1$, λ_t is bounded along the finite reference trajectory, and by (24) and (28), we get $\|r'_t - r_t\| \leq L_t \|r'_{t-1} - r_{t-1}\| + D_t \eta$. The induction hypothesis then gives $\|r'_t - r_t\| \leq B_t \eta$ for a suitable constant B_t . This closes the induction.

Taking the maximum of the finitely many constants over all samples and all steps, we obtain $C_{\text{CCDS}} > 0$ such that $\|r'_{l,T_l} - r_{l,T_l}\| \leq C_{\text{CCDS}}\eta$ for every l . The transported perturbed path is an admissible CCDS realizer f'_{CCDS} on $U' = U$. Therefore $\varepsilon_{\text{param}}(A, U', f'_{\text{CCDS}}; \Xi) = \frac{1}{N} \sum_{l=1}^N \|r'_{l,T_l}\| \leq \frac{1}{N} \sum_{l=1}^N \|r_{l,T_l}\| + \frac{1}{N} \sum_{l=1}^N \|r'_{l,T_l} - r_{l,T_l}\| \leq \varepsilon_{\text{param}}(A, U, f_{\text{CCDS}}; \Xi) + C_{\text{CCDS}}\eta$. Taking $C_{\text{param}} := C_{\text{CCDS}}$ and recalling that $\eta = \|G' - G\|_2$ gives the desired bound.

How to choose δ_0 explicitly: for every $l = 1, \dots, |\Xi_{\text{sub}}|$, define $\delta_{1,l,t} > 0$ such that $\|G' - G\|_2 \leq \delta_{1,l,t}$ implies $\|q'_{l,t} - q_{l,t}\| \leq \nu/2$; and $\delta_{2,l,t} > 0$ such that $C_{m,l,t}(\|r'_{l,t-1} - r_{l,t-1}\| + \delta_{2,l,t}) \leq \eta_0^2/2$. For each $j = 1, 2$, set $\delta_{j,l} := \min_{1 \leq t \leq T_l} \delta_{j,l,t}$ and $\delta_j := \min_{1 \leq l \leq |\Xi_{\text{sub}}|} \delta_{j,l}$. Finally, set $\delta_0 := \min(\gamma/2, \delta_1, \delta_2)$. $\square$

Proof of Theorem 2.9 (Intrinsic local orbit-certificate sparsity of full CCDS). Fix i . Write $p = p_i$, $S = S_i^*$, $E = E_i^*$, $z = z_i^* = \Pi_E p$, $\rho = \rho_i^*$. By definition, $\rho = \|p - z\| \leq \eta$. Since $z \in E$, and E is spanned by exactly the atoms indexed by S , we may write $z = \sum_{r \in S} c_r a_{i,r}$. Define $u := \frac{z}{\|z\|}$. Then $u \in E$, so u is an admissible interior orbit innovation supported on the atoms in S . Start full CCDS from $Y_0 = \{0\}$, $\hat{p}_0 = 0$, $r_0 = p$. The CCDS distance equation at this step is $\rho(r_0 - \lambda u, Y_0) = \eta$. Since $Y_0 = \{0\}$, this becomes $\|p - \lambda u\| = \eta$. Now compute the minimal feasible deviation along u . We have $\langle p, u \rangle = \left\langle z + (p - z), \frac{z}{\|z\|} \right\rangle$. Because $z = \Pi_E p$, the residual $p - z$ is orthogonal to E , and $u \in E$. Hence $\langle p, u \rangle = \left\langle z, \frac{z}{\|z\|} \right\rangle = \|z\|$. Therefore, $d_{\min}^2 = \|p\|^2 - |\langle p, u \rangle|^2 = \|p\|^2 - \|z\|^2$. Since $z = \Pi_E p$, $\|p\|^2 = \|z\|^2 + \|p - z\|^2$. Thus $d_{\min} = \|p - z\| = \rho \leq \eta$. So the CCDS distance equation is feasible. The CCDS root formula gives $\lambda = \|z\| - \sqrt{\eta^2 - \rho^2}$. Now define $\hat{p}_i := \lambda u$. By construction, $\|p_i - \hat{p}_i\| = \eta$. Since $u = \frac{z}{\|z\|} = \frac{1}{\|z\|} \sum_{r \in S} c_r a_{i,r}$, we get $\hat{p}_i = \lambda u = \sum_{r \in S} \frac{\lambda c_r}{\|z\|} a_{i,r}$. Define $\alpha_{i,r}^* := \lambda c_r / \|z\|$ for each r . Therefore, $\text{supp } \alpha_i^* \subseteq S$, so $|\text{supp } \alpha_i^*| \leq |S| = s_i(\eta)$. This proves the theorem. $\square$

A.5 Proofs for prediction

Proof of Theorem 3.2 (Geometric in-sample prediction bound). Fix $i \in \{1, \dots, n\}$. Since $f(x_i) = y_i$, $(\Pi_U f)(x_i) - \tilde{f}(x_i) = ((\Pi_U f)(x_i) - \text{CCDS}_{\mathcal{D}}(y_i)) + (\text{CCDS}_{\mathcal{D}}(y_i) - \tilde{f}(x_i))$. Hence, by the triangle inequality, $\|(\Pi_U f)(x_i) - \tilde{f}(x_i)\|_{\ell^2} \leq \|(\Pi_U f)(x_i) - \text{CCDS}_{\mathcal{D}}(y_i)\|_{\ell^2} + \delta_i$. Averaging over $i = 1, \dots, n$ gives $\|\Pi_U f - \tilde{f}\|_{\Xi_{\text{train}}} \leq \|\Pi_U f - \text{CCDS}_{\mathcal{D}} \circ f\|_{\Xi_{\text{train}}} + \delta$.

Finally, $f - \tilde{f} = (f - \Pi_U f) + (\Pi_U f - \tilde{f})$. Therefore, $\|f - \tilde{f}\|_{\Xi_{\text{train}}} \leq \|f - \Pi_U f\|_{\Xi_{\text{train}}} + \|\Pi_U f - \tilde{f}\|_{\Xi_{\text{train}}}$, and substituting the preceding estimate yields $\|f - \tilde{f}\|_{\Xi_{\text{train}}} \leq \|f - \Pi_U f\|_{\Xi_{\text{train}}} + \|\Pi_U f - \text{CCDS}_{\mathcal{D}} \circ f\|_{\Xi_{\text{train}}} + \delta$. $\square$

Proof of Theorem 3.3 (Geometric out-of-sample prediction bound). Fix $j \in \{1, \dots, m\}$, and choose $(x_{i(j)}, y_{i(j)}) \in \Xi_{\text{train}}$ such that $\|\bar{x}_j - x_{i(j)}\|_{\ell^2} \leq \delta$. Then

$$f(\bar{x}_j) - \tilde{f}(\bar{x}_j) = (f(\bar{x}_j) - \Pi_U f(\bar{x}_j)) + (\Pi_U f(\bar{x}_j) - \Pi_U f(x_{i(j)})) + (\Pi_U f(x_{i(j)}) - \tilde{f}(x_{i(j)})) + (\tilde{f}(x_{i(j)}) - \tilde{f}(\bar{x}_j)).$$

Applying the triangle inequality,

$$\begin{aligned} \|f(\bar{x}_j) - \tilde{f}(\bar{x}_j)\|_{\ell^2} &\leq \|f(\bar{x}_j) - \Pi_U f(\bar{x}_j)\|_{\ell^2} + \|\Pi_U f(\bar{x}_j) - \Pi_U f(x_{i(j)})\|_{\ell^2} \\ &\quad + \|\Pi_U f(x_{i(j)}) - \tilde{f}(x_{i(j)})\|_{\ell^2} + \|\tilde{f}(x_{i(j)}) - \tilde{f}(\bar{x}_j)\|_{\ell^2}. \end{aligned}$$

Since Π_U is an orthogonal projector, $\|\Pi_U\| \leq 1$. Hence, by Assumption 2,

$$\|\Pi_U f(\bar{x}_j) - \Pi_U f(x_{i(j)})\|_{\ell^2} \leq \|f(\bar{x}_j) - f(x_{i(j)})\|_{\ell^2} \leq L_f \delta.$$

By Assumption 3, $\|\tilde{f}(x_{i(j)}) - \tilde{f}(\bar{x}_j)\|_{\ell^2} \leq L_{\tilde{f}} \delta$. Therefore,

$$\|f(\bar{x}_j) - \tilde{f}(\bar{x}_j)\|_{\ell^2} \leq \|f(\bar{x}_j) - \Pi_U f(\bar{x}_j)\|_{\ell^2} + \|\Pi_U f(x_{i(j)}) - \tilde{f}(x_{i(j)})\|_{\ell^2} + (L_f + L_{\tilde{f}})\delta.$$

Averaging over $j = 1, \dots, m$, we obtain

$$\|f - \tilde{f}\|_{\Xi_{\text{test}}} \leq \|f - \Pi_U f\|_{\Xi_{\text{test}}} + \max_{x:(x,y) \in \Xi_{\text{train}}} \|\Pi_U f(x) - \tilde{f}(x)\|_{\ell^2} + (L_f + L_{\tilde{f}})\delta.$$

To conclude, it suffices to apply the triangle inequality to the training term:

$$\|\Pi_U f(x) - \tilde{f}(x)\|_{\ell^2} \leq \|\Pi_U f(x) - \text{CCDS}_{\mathcal{D}}(f(x))\|_{\ell^2} + \|\text{CCDS}_{\mathcal{D}}(f(x)) - \tilde{f}(x)\|_{\ell^2}.$$

This proves the theorem. $\square$

Table 7: Experiment 1: Phase 2 MP/CCDS behavior under deviation-control ablation on the weak-alignment synthetic dataset (7% scaffold budget, $d = 10$, $M = 500$, operator family ILL-SVD).

γ_{dev}	$\varepsilon_{\text{proj}}$	$\varepsilon_{\text{param}}^{\text{fast}}$	atm x samp/Kd mean (fast solv)	Top-3 mass fast solv	$\varepsilon_{\text{param}}^{\text{full CCDS}}$	samples with $\varepsilon_{\text{param}}^{\text{full CCDS}} \leq 10^{-6}$	atm x samp/Kd mean (full CCDS)	Top-3 mass full CCDS
0.0	1.35×10^{-10}	6.36×10^{-8}	9/230	96.8%	2.20×10^{-7}	99.2 %	3/230	100%
0.5	1.35×10^{-10}	6.75×10^{-8}	10/230	95.1 %	5.34×10^{-4}	54.4 %	3/230	100%
0.9	1.35×10^{-10}	3.87×10^{-3}	9/230	91.9%	1.18×10^{-3}	41.2 %	3/230	100%

Table 8: Experiment 2: Interpretability metrics for the fast/full CCDS solver.

Operator	atm x samp/Kd mean (fast solv)	Top-3 mass fast solv	Top Atoms among classes 0-5 (fast solver)	atm x samp/Kd mean (full CCDS)	Top-3 mass full CCDS
I	13/450	88.6%	323 (29–32%), 329 (20–28%), 318(20–24%)	2/450	100%
ORTHO	13/450	88.8%	323 (25–32%), 329 (23–27%), 443 (20–26%)	2/450	100%
ILL_SVD	13/450	88.9%	323 (20–27%), 329 (22–27%), 317 (19–22%)	2/450	100%
LOWRANK	13/450	96.3%	323 (19–31%), 443 (18–24%), 329 (19–25%)	2/450	100%
SPIKED	14/450	86.5%	323 (27–33%), 329 (22–26%), 443 (18–24%)	2/450	100%
SPARSE	13/450	88.6%	323 (27–30%), 329 (23–28%), 317 (20–23%)	2/450	100%
TOEPLITZ	13/450	88.2%	329 (25–30%), 323 (24–27%), 325 (18–24%)	2/450	100%
ADV_SCAFF	13/450	88.3%	323 (26–32%), 329 (20–28%), 325 (21–26%)	2/450	100%

Table 9: Experiment 3: Scaffold-quality diagnostic on the synthetic realizability benchmark. Better scaffold quality, measured through lower projection residual and higher explained label-space energy, correlates with lower parametrization error and higher realizability success. Values are mean $\pm$ standard deviation over runs.

Scaffold type	$\varepsilon_{\text{proj}}$	E_{expl}	$\varepsilon_{\text{param}}$
Selected orbit	$1.5\text{e-}10 \pm 9.9\text{e-}11$	1.00 ± 0.0	$7.39\text{e-}8 \pm 4.8\text{e-}9$
Random orbit	$2.15\text{e-}1 \pm 1.1\text{e-}1$	0.85 ± 0.10	$1.4\text{e-}8 \pm 2.9\text{e-}9$
Gaussian	$2.1\text{e-}1 \pm 1.8\text{e-}1$	0.80 ± 0.28	$1.0\text{e-}3 \pm 1.0\text{e-}3$

Table 10: Ablation 3: Discovery-budget ablation. Gains are strongly front-loaded, indicating early recovery of the relevant orbit geometry and saturation of scaffold quality.

Dataset	Metric	1%	3%	5%	7%	10%
Gas Sensor Drift 3000 train samples $d = 10$	Test Accuracy	0.6761	0.6661	0.6621	0.6413	0.6531
	$\varepsilon_{\text{proj}}$	$1.4\text{e-}10$	$5\text{e-}15$	$2.4\text{e-}15$	$7.3\text{e-}11$	$8\text{e-}11$
	$\varepsilon_{\text{param}}$	$1.1\text{e-}7$	$1.4\text{e-}7$	$9.12\text{e-}8$	$8.8\text{e-}8$	$4.7\text{e-}8$
	Nominal atoms	220	280	340	400	490
	Effect Support	8.88	10.29	10.42	9.95	9.04
	Top-3 mass (%)	97.1	97.9	98	97	98
MNIST 5000 train samples $d = 20$	Test Accuracy	0.9298	0.9321	0.9326	0.9308	0.9305
	$\varepsilon_{\text{proj}}$	$4.3\text{e-}15$	$7.1\text{e-}10$	$5.8\text{e-}15$	$6.1\text{e-}15$	$9.6\text{e-}15$
	$\varepsilon_{\text{param}}$	$1.4\text{e-}7$	$1.4\text{e-}7$	$9.1\text{e-}8$	$1.5\text{e-}7$	$1.7\text{e-}7$
	Nominal atoms	440	540	640	740	880
	Effect Support	19.02	17.95	18.79	18.57	18.58
	Top-3 mass (%)	90.8	90.4	91.8	92.4	92.5

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